

Attention as Investment: Traffic Allocation for Long Tail Creators

Qinlu Hu

Department of Operations, Technology and Innovation Management, Nova School of Business and Economics, Universidade NOVA de Lisboa, qinlu.hu@novasbe.pt

Ao Huang

Department of Business Technology, Miami Herbert Business School, University of Miami, axh1828@miami.edu

Ni (Nina) Huang

Department of Business Technology, Miami Herbert Business School, University of Miami, nhuang@miami.edu

Renyu Zhang

Department of Decisions, Operations and Technology, CUHK Business School, The Chinese University of Hong Kong, philipzhang@cuhk.edu.hk

Problem definition: Short-video platforms allocate scarce viewer attention through recommender systems. Long tail creators are an important future supply pipeline, but developing this pipeline requires exposing viewers to creators with limited performance histories, which can reduce current engagement. The platform’s challenge is therefore not simply whether to support long-tail creators, but how to allocate scarce viewer attention to them and which heterogeneous viewer segments are most conducive to their development.

Methodology/results: We leverage two randomized field experiments and one large-scale observational study with two leading short-video platforms in Asia, and synthesize the evidence in an analytical model. Study 1 ($N = 13.6$ million viewers) shows that reducing long tail content increases average engagement, establishing a short-term engagement cost. Further heterogeneity analyses reveal that this cost is substantially smaller for active viewers than for casual viewers. Study 2 ($N = 960,401$ creators) shows that allocating additional attention to long tail creators increases productivity, follower growth, and content performance. Study 3 ($N = 885,478$ newly registered creators) documents positive associations between a higher early active-viewer exposure share and subsequent productivity, performance, and monetization outcomes, with positive associations also visible in six-month cumulative outcomes. Month-specific estimates are strongest early and attenuate heterogeneously across outcomes.

The analytical model maps these findings into cost, return, and reward-productivity primitives. It shows that platforms should rank traffic sources by current engagement cost per unit of effective creator development, yielding conditional starve, active-first incubate, and broad boost regimes.

Managerial implications: Platforms should view long tail exposure as an attention allocation decision, considering both the amount of attention provided and the types of viewers from whom it is sourced. Rather than uniformly distributing exposure, platforms should allocate attention to viewer segments with lower engagement costs per unit of creator development. In our setting, active viewers serve as a lower-cost incubation channel, enabling creator development while preserving casual viewer engagement.

Key words: Attention Allocation, Recommender Systems, Platform Operations, Field Experiments, Creator Economy

1. Introduction

Short-video platforms such as TikTok, YouTube Shorts and Instagram Reels¹ are among the largest digital businesses in the world. TikTok alone reports over a billion monthly active users, and the global short-video market is estimated at \$45.34 billion in 2024 and projected to reach \$835.69 billion by 2035 (Chaudhary 2025). These platforms are built on recommender systems that continuously allocate viewer attention across creators and content.

Short-video platforms allocate viewer attention billions of times each day through their recommender systems. Viewer attention, however, is a finite resource: a viewer can watch only one video at a time. For these platforms, every allocation decision involves a trade-off: attention given to an established creator is denied attention to a creator in the long tail. Platforms thus face an unavoidable operational problem: how to allocate viewer attention. Configuring operational resources on digital platforms is an active area of empirical research in operations management (Hu and Benjaafar 2020, Li and Netessine 2020). We extend this tradition by studying viewer attention as an operational resource whose allocation shapes both current consumption and future supply development.

Although recommender systems allocate attention efficiently for current engagement, they may not account for the future supply value of emerging creators. They are designed to maximize short-term engagement (Covington et al. 2016, Besbes et al. 2016, Raffeian 2023). To do so, they rely on historical performance data (watch time, likes, shares) to predict which content will generate the most interaction. This creates what the literature calls “popularity bias” (Abdollahpouri et al. 2019): a systematic preference for established creators, who have abundant historical data and whose content the algorithm can recommend with high confidence, against long tail creators, defined throughout this paper as those with fewer than 1,000 followers², who lack such data and receive correspondingly less attention. The result is a self-reinforcing cycle: long tail creators receive little exposure and therefore generate no performance signals, which in turn ensures they continue to receive little exposure. This is also referred to as the “cold-start” problem (Lika et al. 2014, Liang et al. 2024): the algorithm cannot learn what it has never seen. Starved of the initial attention needed to demonstrate their potential, many long tail creators exit the platform. Over time, the cycle concentrates power among a small number of star creators, thins the platform’s content supply pipeline, and exposes the platform to systemic risk from star turnover (Forderer et al. 2025, Qian and Xie 2026).

This tension arises because viewer attention on platforms serves two distinct roles: it generates immediate engagement and, when allocated to long tail creators, provides the feedback needed for

future supply development. Long tail creators therefore represent an important future supply pipeline: the pool from which tomorrow’s successful creators may emerge. A platform cannot sustain itself on its current stars indefinitely, because maintaining a healthy content ecosystem requires continuously developing new sources of supply. Platforms must therefore consider allocating viewer attention to long tail creators as an investment in future supply development.

However, this attention investment is not costless. Exposing viewers to unfamiliar long tail content may reduce current engagement because viewers often prefer familiar, established creators. The platform consequently faces an investment problem: it must incur short-term engagement costs today in exchange for potential long-term creator-development benefits. Prior research documents a consumption-production trade-off from changing recommender exposure to less-popular creators (Zeng et al. 2024) and shows that sustained support for amateur creators can eventually offset initial usage losses through improvements in future content supply (Xia 2025). These findings establish that creator development involves an intertemporal trade-off, but they do not determine how this scarce attention should be sourced and allocated. Because traffic sources may differ in both engagement cost and developmental value, uniform treatment of attention may be inefficient. Viewer heterogeneity may be particularly relevant because active viewers can respond differently to recommendation diversity (Chen et al. 2024a). The operational challenge is therefore not merely whether to develop the long tail, but how much attention the platform should provide and which viewer segments should supply it.

We therefore ask three linked questions: (I) Cost: What is the short-term engagement cost of allocating attention to long tail creators and how does this cost vary across viewer segments? (II) Return: Does algorithmic attention allocation stimulate long tail creator outcomes (i.e., productivity, performance, and growth)? (III) Allocation: Given these costs and returns, how should platforms allocate attention to long-tail creators by leveraging viewer segments with different developmental value?

To answer these questions, we leverage two randomized field experiments and one large-scale observational study conducted in collaboration with two leading short-video platforms in Asia. Study 1, a viewer-side experiment ($N = 13.6$ million), shows that reducing long tail content increases total video watching time by 0.33%, establishing a short-term engagement cost. This cost varies across viewer segments: It is substantially larger among casual than active viewers. Study 2, a creator-side experiment ($N = 960,401$), shows that additional attention increases long tail creators’ video production (+5.87%), production days (+2.89%), and follower growth (+289.81%), establishing a causal return. Study 3 tracks 885,478 newly registered creators on a second platform, examining how active-viewer exposure during the initial 30-day period is associated with subsequent creator development. A higher share of early exposure from active viewers is positively associated with creators’

subsequent productivity, performance, growth, and monetization outcomes, with these associations persisting over time.

The three empirical studies are designed to identify complementary primitives rather than provide interchangeable replications. Study 1 estimates the cost of attention in the currency the platform directly manages, viewer engagement, and shows that this cost varies across viewer segments. Study 2 establishes that the same operational input generates creator-side returns, transforming an apparent short-term sacrifice into a potential intertemporal investment. Study 3 extends the evidence across platforms and time horizons by documenting how subsequent creator outcomes vary with the composition of early attention. Because its observational design does not identify causal effects of attention composition, Study 3 informs heterogeneity in attention productivity rather than estimating a causal allocation effect. Together, these studies provide the empirical primitives needed for a prescriptive attention allocation framework.

Further, we integrate these findings into an analytical model that compares the short-term engagement cost of long tail exposure with its discounted creator-development value. Study 1 identifies the cost side, Study 2 identifies the causal short-term return, and Study 3 motivates heterogeneity in attention productivity. The model separates the decision of how much attention to provide from the decision of how to allocate that attention across viewer segments. It ranks traffic sources by current engagement cost per unit of effective creator development and characterizes conditional allocation policies based on the engagement cost and developmental value of different traffic sources.

This paper makes two linked contributions. First, we empirically decompose attention allocation into its viewer-side cost, creator-side return, and attention-source heterogeneity. Prior research establishes that creator support can involve short-term viewer costs and longer-term supply benefits. We build on this trade-off by showing that the value of long-tail exposure depends not only on how much attention is allocated, but also on who provides that attention.

Second, we develop a segment-aware attention allocation framework that translates empirical primitives into operational decisions. The framework identifies when investing attention in creator development is worthwhile, which viewer segments are most valuable sources of attention, and when broader allocation is justified. Rather than prescribing a universal exposure policy, the framework provides a conditional allocation principle based on the engagement cost and developmental value of different traffic sources.

Empirically, the paper does not directly estimate a platform-wide optimal dose or the causal effect of assigning active rather than casual viewer traffic. Studies 1 and 2 identify reduced-form causal effects at their respective experimental margins, while Study 3 documents conditional associations between early attention composition and subsequent creator outcomes. The analytical model uses these findings to motivate local cost and creator-response inputs and characterizes the allocation rule

that follows under its maintained assumptions. Accordingly, active-first routing is a setting-specific application of the general effective-unit-cost principle rather than a universal causal claim about active viewers.

Our framework contributes to platform operations by treating recommendation traffic as dynamic development capacity. Platforms do not merely match existing content supply with viewer demand. Through attention allocation choices, they influence which creators acquire the signals, audiences, and incentives needed to become future suppliers. This feedback links current feed allocation decisions to the evolution of future content supply. The framework also clarifies why maximizing immediate average engagement can be dynamically incomplete and why maximizing creator exposure without considering traffic-source heterogeneity can be operationally inefficient.

From a practical perspective, we reframe long tail creator development as an attention investment problem rather than a binary choice of whether to support emerging creators. Platforms should evaluate whether the expected creator-development value justifies current engagement costs and, conditional on investing, allocate attention toward traffic sources with lower engagement costs per unit of development value. In our specific empirical setting, active viewers represent a lower-cost incubation channel because they exhibit smaller engagement losses from long tail exposure and their early attention is associated with stronger subsequent creator outcomes. We highlight that our framework also identifies the measurable primitives that platforms should estimate when designing attention allocation policies.

The rest of this paper is organized as follows. Section 2 reviews the relevant literature. We present the three empirical studies in Section 3 and the analytical model in Section 4. Section 5 concludes.

2. Literature Review

Our paper is connected to two streams of research: (a) digital content creation and (b) algorithmic management.

2.1. Digital Content Creation

Research on digital content creation examines the consequences of user-generated content, including reviews (Duan et al. 2008, Zhu and Zhang 2010, Cui et al. 2020), knowledge sharing (Zhang and Zhu 2011), videos (Zhao et al. 2025), and online communities (Greenstein et al. 2021, Zhang et al. 2022). This literature also studies the factors shaping content contribution, including status and differentiation (Goes et al. 2016, Levina and Arriaga 2014), competition (Wu and Zhu 2022), platform design and advertising (Qian and Jain 2024, Cao et al. 2023, Ren 2024), and artificial intelligence (Zhang et al. 2024, 2026), including emerging search interfaces (Bai et al. 2026).

Prior work often focuses on star creators, whose disproportionate value motivates retention and exclusivity strategies (Han and Ravid 2020, Maclean 2021, Carroni et al. 2024). Recent studies further

show that star creator turnover can affect platform content supply and peer creators’ contributions (Forderer et al. 2025, Qian and Xie 2026). However, dependence on a small number of stars may create ecosystem vulnerability (Bhargava 2022).

Long tail creators pose a different development challenge. Although they constitute a large share of the creator population, they individually receive limited attention (Wu et al. 2023). They face a cold-start problem because recommendation systems rely on historical engagement signals that cannot be generated without prior exposure (Lika et al. 2014, Liang et al. 2024). Nevertheless, long tail creators contribute to content variety, replenish future creator supply, and reduce dependence on a small number of suppliers (Kumar et al. 2014, Gaenssle and Budzinski 2021, Cutolo and Kenney 2021).

Platforms have developed various mechanisms to support creators, including revenue sharing, affiliate marketing, symbolic awards, feedback, and social nudges (Jain and Qian 2021, Gu et al. 2024, Gallus 2017, Burtch et al. 2022, Huang et al. 2019, Zeng et al. 2023). However, these mechanisms primarily address incentives after creators obtain visibility. For long tail creators, the fundamental bottleneck is earlier: obtaining the exposure needed to generate performance signals and growth.

This challenge connects creator development to supplier development in operations management (Wan and Wu 2017). An earlier conference study provides initial experimental evidence that platform-controlled traffic allocation can support small creator development while balancing viewer engagement (Hu et al. 2024). The current study extends this foundation by incorporating cross-platform longitudinal evidence and an analytical model that distinguishes the quantity of attention from its allocation across heterogeneous traffic sources.

Related work further documents the tension between short-term consumption and long-term creator supply. Zeng et al. (2024) identify a consumption–production trade-off when changing recommender exposure to popular creators, while Xia (2025) shows that sustained support for amateur creators can eventually offset initial engagement losses through improvements in content quantity, quality, and diversity. Xia (2025) also finds that highly engaged viewers respond more negatively to additional program-promoted amateur content. Our Study 1 examines a different allocation margin: the platform removes long tail content from the recommendation candidate pool while leaving its usual ranking mechanism in place. We find that active viewers experience larger engagement losses, or smaller engagement gains, from this reduction. The contrasting patterns suggest that active viewers may value organically recommended long tail diversity without necessarily being more tolerant of incremental promotional exposure. Accordingly, we treat viewer-segment cost rankings as intervention-specific empirical primitives rather than as universal properties of active and casual viewers.

Building on this literature, we address a distinct operational question. Rather than evaluating whether a given creator-support program generates a positive overall return, we ask how much attention a platform should provide and how that attention should be sourced across heterogeneous viewer segments. We separate the developmental-attention target from its traffic-source allocation and rank traffic sources by their current engagement cost per unit of effective creator-development value.

2.2. Algorithmic Management

Operations management research examines how digital platforms configure resources and matching environments through algorithms (Hu and Benjaafar 2020, Li and Netessine 2020). Recommender systems are central to this process because they determine how viewer attention is allocated across creators and content while predicting engagement outcomes (Bhargava 2022, Covington et al. 2016).

Research has demonstrated both the benefits and limitations of algorithmic management. Recommender systems can improve consumption and matching efficiency (Zhou et al. 2010, Claussen et al. 2019), but algorithmic optimization can also generate biased outcomes, polarization, reduced content diversity, and choice overload (Lambrecht and Tucker 2019, Brown et al. 2022, Anderson et al. 2020, Long et al. 2024). These concerns have motivated technical work on balancing recommendation accuracy and diversity (Kim et al. 2019, Yin et al. 2012) and analytical research on how recommender systems influence creator incentives, entry, quality, and variety (Qian and Jain 2024, Zou et al. 2025). Related empirical studies show that algorithmically mediated operational choices can shape participant behavior through platform incentives and interface design (Xu et al. 2023, 2024).

A related question concerns whether algorithmic allocation should account for participant heterogeneity. Viewers differ in their consumption patterns (Heinonen 2011), and active users may tolerate greater recommendation diversity without similar engagement losses (Chen et al. 2024a). More broadly, heterogeneous responses affect how interventions should be targeted (Gupta et al. 2020), while the value of human–algorithm collaboration varies with experience, skill level, and task complexity (Knight et al. 2026). This observation relates to targeted exploration, where optimal mechanisms selectively expose content to subsets of users (Papanastasiou et al. 2018, Che and Horner 2018). However, such allocation strategies have rarely been tested empirically at platform scale.

Because content platforms are two-sided systems, algorithmic interventions may also create cross-side effects (Johari et al. 2022). Building on this literature, the current study treats attention allocation as an operational lever whose value depends on both the quantity and viewer composition of developmental exposure. We connect the algorithmic management of attention with the broader challenge of balancing current platform performance and long-term ecosystem development.

3. Empirical Studies

3.1. Institutional Contexts

We collaborated with two leading Asian short-video platforms (Partner Platforms A and B), each with hundreds of millions of daily active users.³ Like TikTok, both provide short videos through a scrolling recommendation page personalized using viewers’ interests, histories, and engagement (Figure 1). The technical details of their recommender system appear in Appendix A. As two-sided markets, viewers consume free content across numerous categories, creators pursue engagement and followers, and the platforms earn advertising and sponsorship revenue, part of which is shared with creators based on performance.

Three complementary studies answer our research questions. Study 1 uses a viewer-side randomized experiment to estimate the engagement cost of long tail exposure and its heterogeneity by viewer activeness; Study 2 uses a creator-side randomized experiment to estimate the developmental return to attention; and Study 3 provides observational evidence relating early attention composition to subsequent creator outcomes. Finally, we integrate these inputs into an analytical model to provide insights into the optimal traffic allocation policy in Section 4.

Studies 1 and 2 use a fixed, one-time Platform A data extract that could not subsequently be extended under its data-governance policy. Study 3 uses observational data from Platform B under a more flexible data-sharing arrangement. Appendix B documents the extracts, variables, and descriptive statistics.⁴

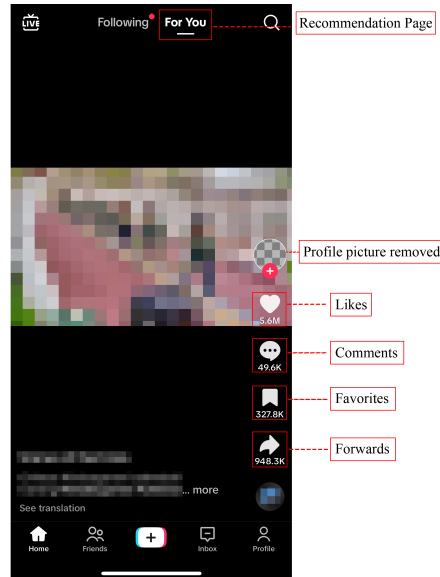


Figure 1: Screenshot of the User Interface on TikTok⁵

3.2. Study 1: Viewer-Side Randomized Field Experiment

3.2.1. Experimental Design To quantify the short-term engagement cost of long tail content, Platform A conducted a viewer-side randomized field experiment on December 2–11, 2022. Viewers

were assigned to a treatment group, for which 50% of videos from long tail creators (those with fewer than 1,000 followers) were removed from the recommendation candidate pool, or an untreated control group.⁶ The intervention altered eligible video supply rather than the recommender system itself (Kellogg et al. 2020). Table 1 summarizes the design.

Table 1: Experimental Design in Study 1

Group	Manipulation	Observations
Treatment	Remove 50% of long tail creators’ video supply from the candidate video pool of each viewer	4 532 431
Control	N/A	9 065 114

Notes: Observations correspond to unique viewers included in the experiment. Randomization is conducted at the viewer level.

3.2.2. Data and Randomization Check We obtained viewer demographic and engagement data for all 13,597,545 viewers who participated in the randomized field experiment in Study 1 (4,532,431 in the treatment group and 9,065,114 in the control group). Variable definitions and descriptive statistics are reported in Appendix B. To protect data confidentiality, the partner platform multiplied each continuous variable by a random, undisclosed constant before sharing the data. Consequently, all numerical values represent scaled metrics, and coefficients should be interpreted in relative terms. Furthermore, the measurements used in the descriptive statistics and regression analyses were independently rescaled by different positive numbers, so they should not be compared directly.

We first verify that randomization produced comparable groups. As shown in Appendix C, treatment and control groups are balanced on all pre-experiment demographics and engagement metrics (all $p > 0.05$). We next confirm that the manipulation operated as intended. As reported in Table 2, the treatment significantly reduced the share of long tail content in treated viewers’ feeds by 13.21% ($p < 0.01$) without altering the total number of videos viewed, confirming that the manipulation selectively reduced long tail exposure rather than changing overall content consumption.

Table 2: Manipulation Check in Study 1

Variable	<i>NumViews</i>	<i>LongTailViewShare</i>
<i>Treatment</i>	0.549 (0.519)	−0.014*** (0.0001)
<i>Relative Effect Size (%)</i>	0.16	−13.21***
Observations	13 597 545	13 597 545

Notes: Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3.2.3. Estimation Equations Because treatment varied long tail exposure at the viewer level, we estimate its effects on viewer engagement using the OLS specification in Equation (1):

$$Y_i = \alpha + \beta \cdot \text{Treatment}_i + \varepsilon_i \quad (1)$$

where Treatment_i is a binary indicator for whether viewer i is in the treatment group (where 50% of long tail creators' videos were removed from the recommendation pool) versus the control group. Y_i represents our engagement metrics of interest, measured at the viewer level during the experiment period. We focus primarily on consumption-based engagement outcomes, including total video watching time (*WatchingTime*) and complete views (*NumCompleteViews*), which capture the overall depth of viewer engagement on the platform. We additionally examine interaction-based engagement outcomes, including likes (*NumLikes*), comments (*NumComments*), and forwards (*NumForwards*), as supporting measures of viewer engagement.

We next test whether treatment responses differ between active and casual viewers, consistent with evidence that platform behavior varies with user activeness (Heinonen 2011), using Equation (2):

$$Y_i = \alpha + \beta_1 \cdot \text{Treatment}_i + \beta_2 \cdot \text{Active}_i + \beta_3 \cdot \text{Treatment}_i \times \text{Active}_i + \varepsilon_i \quad (2)$$

where Active_i is a binary label for viewer i , equal to one if the viewer's total number of video views during the week prior to the experiment (November 25–December 1, 2022) exceeds a platform-defined threshold, and zero otherwise.⁷ The coefficient β_3 allows us to test whether active (vs. casual) viewers respond differently to changes in long tail creator content.

3.2.4. Results We first report the main results of Study 1 in Table 3. The viewers in the treatment group (where 50% of long tail creators' videos were removed from the recommendation pool) significantly increased their total video watching time by 0.33% ($\hat{\beta} = 24.906$, $p < 0.05$) and number of complete views by 0.44% ($\hat{\beta} = 0.585$, $p < 0.01$), respectively, compared to the control group. The effects on other engagement indicators are also positive: the number of likes increased by 0.75% ($\hat{\beta} = 0.017$, $p < 0.05$), while the effects on comments and forwards are statistically insignificant. Because treatment reduced the eligible long-tail supply available to the recommender, the positive average effects identify a short-run engagement cost of baseline long-tail availability at this experimental margin.⁸

Next, we present heterogeneous treatment effects by viewer activeness in Table 4. As a sanity check, the direct effects of the active indicator ($\hat{\beta}_2$) are positive and significant across all outcomes, indicating that active viewers are substantially more engaged than casual viewers in general. Meanwhile, the interaction terms between treatment and active viewer status are negative and significant for watching time ($\hat{\beta}_3 = -86.045$, $p < 0.05$) and the number of complete views ($\hat{\beta}_3 = -2.480$, $p < 0.05$). These

Table 3: Main Effects of Removing Long Tail Creator Content on Viewer Engagement in Study 1

Variable	(1)	(2)	(3)	(4)	(5)
	<i>WatchingTime</i>	<i>NumCompleteViews</i>	<i>NumLikes</i>	<i>NumComments</i>	<i>NumForwards</i>
<i>Treatment</i>	24.906** (11.216)	0.585*** (0.190)	0.017** (0.008)	0.001 (0.002)	0.002 (0.004)
<i>Relative Effect Size (%)</i>	0.33**	0.44***	0.75**	0	0.19
Observations	13 597 545	13 597 545	13 597 545	13 597 545	13 597 545

Notes: (a) Partner Platform A multiplied each variable by a random, undisclosed constant before sharing the data. Therefore, the coefficients represent scaled metrics rather than absolute values, and should be interpreted in relative terms. (b) Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

estimates indicate that active viewers experience larger engagement losses, or smaller engagement gains, when long tail content is removed from the recommendation candidate pool. In contrast, the interaction effect for comments is positive ($\hat{\beta}_3 = 0.051$, $p < 0.01$), suggesting that some interaction behaviors respond differently from consumption-based engagement measures. Because our primary outcomes capture overall viewer consumption through watching time and complete views, we interpret the negative interaction effects on these outcomes as evidence that active viewers derive greater engagement value from the availability of long tail content than casual viewers do under the platform’s baseline recommendation environment.

Table 4: Heterogeneous Treatment Effects by Viewer Activeness in Study 1

Variable	(1)	(2)	(3)	(4)	(5)
	<i>WatchingTime</i>	<i>NumCompleteViews</i>	<i>NumLikes</i>	<i>NumComments</i>	<i>NumForwards</i>
<i>Treatment</i>	24.691*** (8.831)	0.585*** (0.190)	0.019** (0.008)	−0.002 (0.002)	0.003 (0.004)
<i>Active</i>	58 431.645*** (24.597)	1200.000*** (0.991)	10.020*** (0.021)	1.831*** (0.005)	6.455*** (0.011)
<i>Treatment</i> $\times$ <i>Active</i>	−86.045** (42.588)	−2.480** (1.037)	−0.049 (0.037)	0.051*** (0.009)	−0.008 (0.019)
Observations	13 597 545	13 597 545	13 597 545	13 597 545	13 597 545

Notes: Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Further, interviews with product managers and data scientists of Platform A suggested two possible explanations: long tail content may have lower average perceived quality but greater topical diversity. Using a separate cross-sectional dataset of 188,374,718 creators, we compare long tail creators with other creators. Long tail creators have lower complete view rate (15.70% versus 36.02%) and lower

topic HHI (0.10 versus 0.15; both $p < 0.01$). The lower complete view rate serves as a proxy for lower perceived content quality, whereas the lower HHI indicates broader topic coverage (Cheng and Nault 2012). These descriptive patterns are consistent with the proposed quality and diversity explanations.⁹

To summarize, Study 1 shows that reducing the supply of long tail content increases viewer engagement on average, but that this response is substantially smaller among active viewers. These estimates identify a local cost ordering around the platform’s baseline recommendation environment: the engagement cost of long tail exposure is lower for active viewers than for casual viewers. In the analytical model (Section 4), this experimentally motivated engagement cost ordering is represented as $c_A < c_C$. Study 1 supplies the cost side of the attention-investment problem, but it does not establish whether paying that cost benefits creators. Study 2 (Section 3.3) addresses this question by experimentally allocating additional viewer attention to long tail creators.

3.3. Study 2: Creator-Side Randomized Field Experiment

3.3.1. Experimental Design To estimate the creator response to viewer attention, Platform A conducted a creator-side randomized field experiment from March 18 to April 1, 2022. Long tail creators in the experimental sample were randomly assigned to either a treatment group that received additional viewer exposure through increased content-recommendation weights or a control group that remained unchanged. Table 5 summarizes the design.

Table 5: Experimental Design in Study 2

Group	Manipulation	Observations
Treatment	Allocate additional viewer attention to long tail creators	480 048
Control	N/A	480 353

Notes: Observations correspond to unique creators included in the experiment. Randomization is conducted at the creator level.

3.3.2. Data and Randomization Check We obtained creator-level data for all 960,401 long tail creators (480,048 in the treatment group and 480,353 in the control group) who participated in the randomized field experiment in Study 2. The dataset includes creator-level information on demographics, content productivity, performance metrics, and follower growth within the experiment window. Variable definitions and descriptive statistics are reported in Appendix B. Also, to protect data confidentiality, the partner platform multiplied each continuous variable by a random, undisclosed constant before sharing the data. Consequently, all numerical values represent scaled metrics, and coefficients should be interpreted in relative terms.

We first verify the success of randomization. As shown in Appendix C, treatment and control creators are balanced on all pre-experiment demographics, productivity, performance, and growth

metrics (all $p > 0.05$). We next confirm that the intervention operated as intended. As reported in Table 6, the treatment significantly increased the number of views received by treated creators by 265.65% ($p < 0.01$), confirming that enhanced weighting in the recommender system successfully increased the exposure of long tail creators' content.

Table 6: Manipulation Check in Study 2

Variable	<i>NumViews</i>
<i>Treatment</i>	13 348.583*** (2079.236)
<i>Relative Effect Size (%)</i>	265.65***
Observations	960 401

Notes: Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3.3.3. Estimation Equation In Study 2, we seek to examine the impact of viewer attention allocation on long tail creators' content productivity, performance, and growth outcomes, estimating Equation (3) with OLS regressions:

$$Y_i = \alpha + \beta \cdot \text{Treatment}_i + \varepsilon_i \quad (3)$$

The key independent variable is the treatment indicator, Treatment_i , which equals 1 if a long tail creator belongs to the treatment group and 0 otherwise. The outcomes capture creator productivity (uploaded videos, produced video length, and production days), creator content performance (likes, comments, forwards, and complete views), and creator growth (net new followers and monetization eligibility). Because the intervention allocates additional viewer attention at the creator level rather than the individual-production level, we interpret these outcomes as creator-level responses to increased exposure, with productivity and performance reflecting intermediate development responses and growth capturing broader creator development outcomes.

3.3.4. Results Table 7 reports the Study 2 treatment effects. Table 7, Panel A shows that treated long tail creators upload 5.87% more videos ($\hat{\beta} = 0.174$, $p < 0.01$), produce on 2.89% more days ($\hat{\beta} = 0.087$, $p < 0.01$), and increase total produced video length by 4.02% ($\hat{\beta} = 41.170$, $p < 0.05$). These estimates indicate a broader increase in content-production activity. Additional analyses provide no evidence that increased production is accompanied by deterioration in observed content quality measures (Appendix Table D4). Table 7, Panel B reports large increases in content performance: complete views by 345.89%, likes rise by 506.05%, comments by 190.07%, and forwards by 343.87% (all $p < 0.01$). These outcomes capture realized reach and engagement. Panel C of Table 7 shows that treatment increases net new followers by 289.81% ($\hat{\beta} = 11.334$, $p < 0.01$), whereas the effect on monetization eligibility is statistically insignificant across OLS, Probit, and Logit specifications. Thus, additional attention produces immediate gains in productivity, performance, and

follower accumulation, but the follower gain does not translate into monetization within the experiment window.

Table 7: Effects of Viewer Attention Allocation on Creator Outcomes in Study 2

Panel A: Creator Productivity				
Variable	(1)	(2)	(3)	
	<i>NumUploadVideos</i>	<i>ProducedVideoLength</i>	<i>DaysOfProduction</i>	
<i>Treatment</i>	0.174*** (0.036)	41.170** (17.330)	0.087*** (0.013)	
<i>Relative Effect Size (%)</i>	5.87***	4.02**	2.89***	
Observations	960 401	960 401	960 401	
Panel B: Creator Performance				
Variable	(1)	(2)	(3)	(4)
	<i>NumCompleteViews</i>	<i>NumLikes</i>	<i>NumComments</i>	<i>NumForwards</i>
<i>Treatment</i>	2517.643*** (678.730)	156.702*** (56.408)	3.591*** (1.112)	43.414*** (15.663)
<i>Relative Effect Size (%)</i>	345.89***	506.05***	190.07***	343.87***
Observations	960 401	960 401	960 401	960 401
Panel C: Creator Growth				
Variable	(1)	(2)	(3)	(4)
	<i>NumNewFollowers</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>
<i>Treatment</i>	11.334*** (3.207)	0.0004 (0.002)	0.009 (0.006)	0.024 (0.016)
<i>Relative Effect Size (%)</i>	289.81***	2.43	—	—
Observations	960 401	960 401	960 401	960 401
<i>Estimation</i>	OLS	OLS	Probit	Logit

Notes. Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Altogether, Study 2 demonstrates that allocating additional viewer attention to long tail creators substantially increases their content productivity, performance, and early-stage growth.¹⁰ In the analytical model, this evidence supplies the return primitive: viewer attention can build effective creator capital, so long tail exposure is not merely a current engagement cost but also a productive investment. Meanwhile, viewer attention allocation appears to have insignificant effects on long tail creators' monetization probability during the relatively short duration of the field experiment, which is a key indicator of long-term creator success. This raises meaningful follow-up questions: Are early developmental gains reflected in longer-window outcomes, and are those outcomes associated with the composition of early attention? The heterogeneity analyses in Study 1 indicate that, around the platform's baseline recommendation allocation, removing long tail content is less costly among active than casual viewers. Building on this setting-specific cost ordering, Study 3 uses data from Partner Platform B to examine whether the composition of early attention is also associated with subsequent creator outcomes.

3.4. Study 3: Observational Analysis of Early Attention Composition

3.4.1. Empirical Design In Study 3, we leverage an observational design on Partner Platform B,¹¹ a leading online short video platform in Asia with a substantial user base and features similar to Partner Platform A. The platform implemented a system-level viewer attention allocation policy for all newly registered creators during an initial 30-day exposure period following account activation. Focusing on newly registered creators allows us to study creators at the tail of the follower distribution, which is consistent with our operational definition of long tail creators as those with fewer than 1,000 followers in Studies 1 and 2. The platform’s baseline attention allocation policy guarantees a minimum level of content exposure for these creators. Through consultation with Partner Platform B’s engineering team, we confirmed that the composition of viewer attention is shaped by consistent system-level allocation rules rather than by creator-specific targeting rules. This is consistent with prior literature documenting that recommender systems struggle to match content with appropriate audiences for new creators due to data scarcity (Yin et al. 2012). More broadly, prior work has shown that when platforms lack access to personal-level data, recommendation algorithms become less precise in matching content to users, altering platform outcomes (Sun et al. 2024). Consequently, this early viewer allocation provides useful observational variation in the composition of viewership. To assess observable creator selection, we conduct balance tests using creator covariates available at registration. The balance checks do not indicate severe observable imbalance across the observed creator characteristics, reducing concerns that early variation in active-viewer exposure share is driven by observable creator selection (Appendix C). We therefore interpret Study 3 as complementary observational evidence on the association between early active-viewer exposure share and subsequent creator outcomes, both short-term and long-term.

3.4.2. Data and Measures We obtained data from a random sample of 885,478 creators who registered on Partner Platform B between January 1, 2025, and January 14, 2025. For each creator, we compiled profile and behavioral data beginning from their registration date and extending through the subsequent observation period. Our independent variable, *ActiveViewRatio*, measures the proportion of views from active viewers relative to total views during an initial 30-day exposure period following each creator’s registration. As in Study 1, active-viewer status is a binary label defined at the viewer level: a viewer is labeled active if the viewer’s total number of video views during the past week exceeds a platform-defined threshold, and is labeled casual otherwise. We examine associations with three sets of outcomes: creator productivity, creator performance, and creator growth, both short-term and long-term. To ensure temporal separation between the independent variable and all dependent variables, short-term outcomes are measured during post-exposure month 1, immediately after the initial exposure period used to construct *ActiveViewRatio*. These include productivity

metrics (*NumUploadVideos*, *ProducedVideoLength*, *DaysOfProduction*), performance metrics (*NumLikes*, *NumForwards*, *NumComments*, *NumCompleteViews*), and growth metrics (*NumNewFollowers*, *MonetizeProbability*). Long-term outcomes aggregate the same set of variables over post-exposure months 1–6. Thus, neither outcome window overlaps with the initial exposure window.¹² We also control for creators’ demographic and geographic characteristics (*Gender*, *CityLevel*, *Region*, and *CommunityType*).¹³ Variable definitions and descriptive statistics are reported in Appendix B.

3.4.3. Estimation Equation We employ OLS regression following the specification in Equation (4). *ActiveViewRatio_i* represents the proportion of views from active viewers for creator *i* during the initial 30-day exposure period following registration (ranging from 0 to 1). *Y_i* denotes the outcome variables of interest: short-term outcomes are measured during post-exposure month 1, and long-term outcomes are aggregated over post-exposure months 1–6. Thus, neither outcome window overlaps with the initial exposure period used to construct *ActiveViewRatio*. **X_i** includes the full set of demographic and geographic control variables, namely gender, city level, region, and community type, to account for baseline heterogeneity across creators. The coefficient β captures the conditional association between active-viewer exposure share during the initial exposure period and subsequent creator outcomes; because *ActiveViewRatio* is not randomly assigned, β is not interpreted as a causal effect. We also report specification checks that re-estimate the models without control variables and with log-transformed outcomes in Appendix E. These checks yield broadly similar coefficient patterns.

$$Y_i = \alpha + \beta \cdot \text{ActiveViewRatio}_i + \gamma \cdot \mathbf{X}_i + \varepsilon_i \quad (4)$$

Common system-level allocation rules do not make realized viewer composition exogenous, because creator content and subsequent algorithmic matching can jointly influence the active-viewer share. We therefore interpret *ActiveViewRatio* as a predictive composition measure rather than an externally assigned treatment.

3.4.4. Results

Table 8: Short-Term Associations Between Active-Viewer Exposure and Creator Outcomes in Study 3

Panel A: Creator Productivity				
Variable	(1)	(2)	(3)	
	<i>NumUploadVideos</i>	<i>ProducedVideoLength</i>	<i>DaysOfProduction</i>	
<i>ActiveViewRatio</i>	8.581*** (0.363)	442.123*** (25.845)	1.033*** (0.016)	
<i>Controls</i>	Yes	Yes	Yes	
Observations	885 478	885 478	885 478	
Panel B: Creator Performance				
Variable	(1)	(2)	(3)	(4)
	<i>NumCompleteViews</i>	<i>NumLikes</i>	<i>NumComments</i>	<i>NumForwards</i>
<i>ActiveViewRatio</i>	39.728*** (1.613)	18.822*** (0.622)	5.552*** (0.241)	1.564*** (0.210)
<i>Controls</i>	Yes	Yes	Yes	Yes
Observations	885 478	885 478	885 478	885 478
Panel C: Creator Growth				
Variable	(1)	(2)	(3)	(4)
	<i>NumNewFollowers</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>
<i>ActiveViewRatio</i>	12.937*** (0.532)	0.0003*** (0.0001)	0.644*** (0.112)	2.263*** (0.412)
<i>Controls</i>	Yes	Yes	Yes	Yes
Observations	885 478	885 478	885 478	885 478
<i>Estimation</i>	OLS	OLS	Probit	Logit

Notes. *ActiveViewRatio* is measured during the initial 30-day exposure period following registration; dependent variables are measured during post-exposure month 1. The exposure and outcome windows do not overlap. All specifications include gender, city level, region, and community type controls. Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panels A–C of Table 8 report short-term results of post-exposure month 1. Higher early active-viewer exposure share is associated with higher creator productivity, including number of videos uploaded ($\hat{\beta} = 8.581$, $p < 0.01$), total video length ($\hat{\beta} = 442.123$, $p < 0.01$), and days of production ($\hat{\beta} = 1.033$, $p < 0.01$). The associations extend to performance metrics, including number of complete views ($\hat{\beta} = 39.728$, $p < 0.01$), likes ($\hat{\beta} = 18.822$, $p < 0.01$), comments ($\hat{\beta} = 5.552$, $p < 0.01$), and forwards ($\hat{\beta} = 1.564$, $p < 0.01$). Growth outcomes show a similar pattern, including number of new followers ($\hat{\beta} = 12.937$, $p < 0.01$) and monetization probability ($\hat{\beta} = 0.0003$, $p < 0.01$). The monetization association is also positive in the Probit and Logit specifications reported in the same table. Overall, these short-term patterns are consistent with the experimental findings from Study 2, and motivate us to examine whether active-viewer attention may have higher developmental productivity in the analytical model.

Table 9: Long-Term Associations Between Active-Viewer Exposure and Creator Outcomes in Study 3

Panel A: Creator Productivity				
Variable	(1)	(2)	(3)	
	<i>NumUploadVideos</i>	<i>ProducedVideoLength</i>	<i>DaysOfProduction</i>	
<i>ActiveViewRatio</i>	21.688*** (0.936)	1086.129*** (55.569)	3.486*** (0.067)	
<i>Controls</i>	Yes	Yes	Yes	
Observations	885 478	885 478	885 478	
Panel B: Creator Performance				
Variable	(1)	(2)	(3)	(4)
	<i>NumCompleteViews</i>	<i>NumLikes</i>	<i>NumComments</i>	<i>NumForwards</i>
<i>ActiveViewRatio</i>	107.236*** (4.097)	57.301*** (2.884)	17.818*** (1.148)	4.553*** (1.043)
<i>Controls</i>	Yes	Yes	Yes	Yes
Observations	885 478	885 478	885 478	885 478
Panel C: Creator Growth				
Variable	(1)	(2)	(3)	(4)
	<i>NumNewFollowers</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>
<i>ActiveViewRatio</i>	9.729*** (1.195)	0.001*** (0.0001)	0.312*** (0.060)	1.037*** (0.210)
<i>Controls</i>	Yes	Yes	Yes	Yes
Observations	885 478	885 478	885 478	885 478
<i>Estimation</i>	OLS	OLS	Probit	Logit

Notes. *ActiveViewRatio* is measured during the initial 30-day exposure period following registration; dependent variables are aggregated over post-exposure months 1–6. The exposure and outcome windows do not overlap. All specifications include gender, city level, region, and community type controls. Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9, Panels A–C, reports long-term outcomes accumulated over post-exposure months 1–6. The positive associations with early active-viewer exposure share are also visible over this longer window. For productivity, higher early active-viewer exposure share is associated with more videos uploaded ($\hat{\beta} = 21.688$, $p < 0.01$), greater total video length ($\hat{\beta} = 1086.129$, $p < 0.01$), and more days of production ($\hat{\beta} = 3.486$, $p < 0.01$). For performance, the associations remain positive for number of complete views ($\hat{\beta} = 107.236$, $p < 0.01$), likes ($\hat{\beta} = 57.301$, $p < 0.01$), comments ($\hat{\beta} = 17.818$, $p < 0.01$), and forwards ($\hat{\beta} = 4.553$, $p < 0.01$). For growth, higher early active-viewer exposure share is associated with more new followers ($\hat{\beta} = 9.729$, $p < 0.01$) and higher monetization probability ($\hat{\beta} = 0.001$, $p < 0.01$). This pattern suggests a continuing association over the observation window.

3.4.5. Long-Term Dynamic Associations To examine how these associations evolve over time, we re-estimate the specification in Equation (4) separately for post-exposure months 1–6, replacing Y_i with the corresponding monthly outcome while keeping *ActiveViewRatio* and the control variables unchanged. We then plot the estimated coefficients and their confidence intervals to show how the association with early active-viewer exposure share changes over time. Figure 2 presents

three representative outcomes covering creator productivity, performance, and growth; the remaining outcome-specific plots are reported in Appendix F.

The coefficient for the number of uploaded videos is largest in post-exposure month 1 and declines thereafter while remaining positive and statistically significant through post-exposure month 6. The coefficient for complete views follows a similar pattern, remaining positive and significant throughout the outcome window. The positive association with net new follower acquisition is concentrated in post-exposure months 1–2 and then changes sign or fluctuates around zero, suggesting that the observed follower relationship is front-loaded.

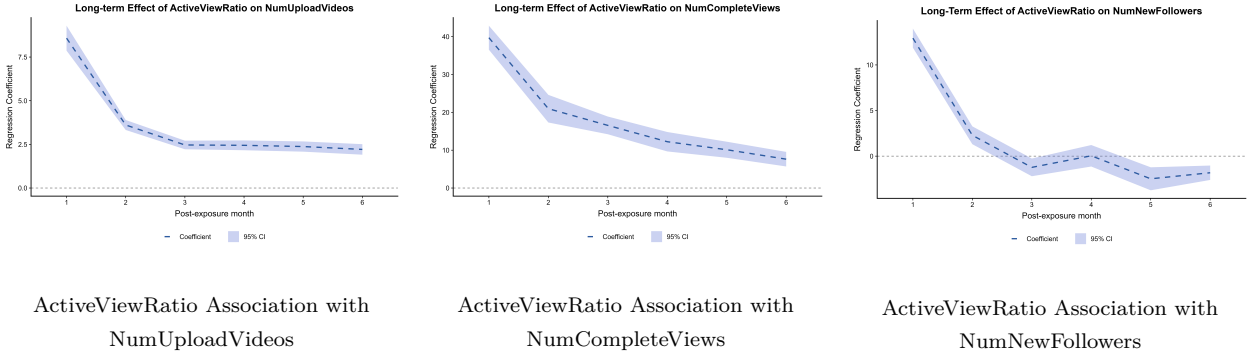


Figure 2: Representative Long-term Dynamic Associations in Study 3

Notes. The dashed lines plot the estimated coefficients on *ActiveViewRatio* from month-specific versions of Equation (4), in which the dependent variable is replaced by the indicated outcome measured separately in each post-exposure month. The horizontal axis indexes post-exposure months 1–6, with post-exposure month 1 denoting the first month immediately following the initial exposure period. The shaded areas report 95% confidence intervals based on robust standard errors. All specifications control for gender, city level, region, and community type ($N = 885,478$ in each regression).

Overall, the coefficient plots show that the associations with active-viewer exposure share during the initial exposure period are strongest immediately after the exposure window and attenuate heterogeneously across outcomes. A suggestive mediation analysis examining patterns consistent with plausible engagement pathways is reported in Appendix G. For the analytical model, Study 3 motivates heterogeneity in the reward-productivity parameter η_j : active-viewer attention may carry more useful engagement signals and may be associated with stronger creator-development responses than casual-viewer attention. Together, Study 1’s lower active-viewer cost and Study 3’s conditional associations motivate examining an active-first ranking. The ranking itself follows from the model’s explicit cost and reward-productivity primitives.

4. Integrating the Evidence: A Model of Attention as Investment

The empirical studies identify complementary pieces of the platform’s decision problem, but no individual study determines how those pieces should be combined. We therefore construct a parsimonious model whose purpose is analytical synthesis. The model does not reconstruct or replace the platform’s recommender system, nor does it provide a structurally estimated optimal deployment policy.

Instead, it asks what prescriptive allocation insights follow under an explicit representation of the viewer-side costs and creator-side developmental responses documented in the empirical studies.

This section combines the empirical primitives in an infinite-horizon allocation model. We derive the operating rule implied by the model under empirically motivated assumptions: the platform decides whether, how much, and through whom to invest in long tail development. Conditional on investing, it routes traffic in increasing order of current engagement cost per unit of effective creator-development value. Our infinite-horizon formulation treats attention as a recurrent investment: the platform repeatedly develops a refreshed long tail pool, and those upgrades help replenish the effective established creator stock as incumbents churn. Under empirically motivated cost and productivity orderings, the model characterizes a traffic-source ranking based on current engagement cost per unit of effective creator-development value. All proofs and detailed derivations, together with the micro-foundations of the model, are relegated to Appendix H.

Table 10: Empirical Evidence, Maintained Mapping, and Model Implications

	Empirical Finding	Model Primitive	Implication
Study 1 main effect	Identifying a positive finite-change engagement penalty around the experimental allocation	$\hat{c} + \frac{\gamma}{2}(2-d)x^u > 0$, where $\hat{c} := \frac{\Theta_A c_A + \Theta_C c_C}{\Theta_A + \Theta_C}$	Long tail exposure has engagement cost.
Study 1 heterogeneity	Motivating a local cost ordering under common first-stage assumptions	$c_A < c_C$	Active viewers are lower-cost recipients of long tail content.
Study 2	Identifying positive short-run creator-side responses	$g_Z > 0$	Attention can build established creator capital.
Study 3	Documenting conditional composition associations	$\eta_A \geq \eta_C$	Active-viewer attention may have higher developmental productivity.

Notes: The notations in this table are defined throughout Section 4.

4.1. Model Setup

We consider an infinite-horizon discount model with time index $t = 0, 1, 2, \dots$. The platform seeks to maximize the total discounted viewer engagement with discount factor $\beta \in (0, 1)$. There are two viewer segments $j \in \{A, C\}$, *active* and *casual*, with stationary attention endowments (feed slots per period) Θ_A and $\Theta_C > 0$, respectively. Let $\Theta := \Theta_A + \Theta_C$. Stationarity is justified in our setting: the total traffic of each segment is governed by viewers' time budgets and is roughly stationary for a large-scale UGC platform. Empirically, Study 1's manipulation check shows that varying the long tail share leaves the total number of videos viewed unchanged (Table 2), i.e., attention supply does

not respond to the traffic allocation. Consistent with our context, the active segment is a small share of viewers commanding a disproportionately high share of attention (Endnote 7), so Θ_A is of the same order as Θ_C .

The state variable is effective established creator capital in period t , denoted by $N_{K,t}$. This continuous state summarizes the production, audience, and performance capacity of creators who have moved beyond the cold-start stage. It need not be interpreted as an integer count of star creators. In each period, the platform has access to a refreshed breadth N_L of long tail creators available for development. Equivalently, $N_{L,t} = N_L$ for all t . The identity of the long tail creators can turn over across periods. The assumption fixes the scale and breadth of the available long tail pool. On large-scale UGC platforms, low-follower creators continuously enter while inactive or unsuccessful creators quickly stop producing, so the relevant developmental long tail pool is naturally replenished. The key question of this section is how the platform should route scarce viewer attention conditional on a long tail pool of a given depth, rather than how creator entry itself is generated. This approach keeps the dynamic state space one-dimensional while allowing pool breadth to affect exposure congestion and creator development.

The platform’s decision variables are the feed shares $x_{j,t} \in [0, \bar{x}]$, where $0 < \bar{x} < 1$: the fraction of segment j ’s period- t traffic filled with long tail content. We use a common cap to focus on routing economics rather than segment-specific capacity constraints. Viewer traffic allocated to the long tail is automatically denied to established content.

Engagement. A segment- j viewer’s per-slot engagement from a feed with long tail share x , given creator pools (N_K, N_L) , is

$$\pi_j(x; N_K, N_L) = v(N_K) - c_j(N_K)x - \frac{\gamma(N_L)}{2}x^2, \quad c_j(N_K) = \Delta(N_K) - b_j, \quad \gamma(N_L) = \frac{\bar{\gamma}}{m(N_L)}, \quad (5)$$

where $\Delta(N_K) := v(N_K) - v_L$ is the quality gap between established and long tail content and $b_A > b_C \geq 0$ is segment j ’s taste for the diversity that long tail content supplies. Three features matter. First, the baseline $v(N_K)$ is engagement per slot allocated to established content. Second, the linear “tax” $c_j(N_K)$ on long tail exposure is the quality gap net of diversity taste. Study 1’s descriptive evidence motivates the quality-gap and diversity components, while its heterogeneous treatment effects motivate the ordering $c_A < c_C$, around the baseline recommendation allocation studied in the experiment. Because $\Delta(\cdot)$ increases in N_K , a larger established pool raises the opportunity cost of every long tail slot. Third, the satiation term $\gamma(N_L)$ falls with the size of the refreshed long tail pool N_L : the marginal long tail slot duplicates topics more slowly when the tail is broader. Specifically, $\bar{\gamma} > 0$ is the baseline satiation or congestion penalty from showing more long tail content in a viewer’s feed and $m(N_L)$ is the effective variety or niche coverage supplied by a refreshed long tail pool of size N_L . Appendix H.1 derives (5) from viewer utility.

Development. Let $\eta_j > 0$ denote the developmental productivity of one unit of segment- j attention, and let

$$Z_t = \eta_A \Theta_A x_{A,t} + \eta_C \Theta_C x_{C,t} \quad (6)$$

denote effective attention routed to the long tail in period t . The η_j 's are reward-side parameters, not attention endowments: Θ_j scales the current engagement consequences of filling feed slots, whereas $\eta_j \Theta_j$ measures the creator-development value of those slots. The causally conservative baseline sets $\eta_A = \eta_C = 1$, so active-first routing is driven by the lower active-viewer engagement cost in Study 1. The heterogeneous-productivity extension allows $\eta_A \geq \eta_C > 0$, motivated by Study 3's observational finding that higher early active-viewer share is associated with stronger subsequent creator outcomes. Because Study 3 is correlational, we treat $\eta_A \geq \eta_C$ as a sensitivity-motivated extension, not as an identified causal estimate.

The platform cannot rank latent long tail creator ability ex ante, so effective attention is spread across the refreshed tail, giving each creator the developmental dose $z_t = Z_t/N_L$. Following Study 2, and consistent with the associations in Study 3, the dose develops creator capital, and the effective increment

$$g(Z, N_L) = N_L H(Z/N_L), \quad (7)$$

contributes to effective established-content supply in the next period, replacing established creator capital that depreciates at rate $\delta \in (0, 1]$, i.e.,

$$N_{K,t+1} = (1 - \delta)N_{K,t} + g(Z_t, N_L). \quad (8)$$

The function $H(\cdot)$ is increasing and strictly concave on the feasible dose range, with $H(0) = 0$. The model captures the production, audience growth, and performance improvements documented in Study 2. It need not imply that treated creators immediately become star creators. The perspective form (7), which follows from creator anonymity and replication, has two structural implications (Appendix H.1): effective attention and tail breadth are complements (i.e., $\partial^2 g / \partial Z \partial N_L > 0$), and a broader tail yields more developed creator capital at any effective-attention level. We also emphasize that Study 2 identifies a positive short-run response to attention. The conversion of those responses into next-period effective creator capital is a maintained reduced-form transition, as Equation (8), used to study the intertemporal allocation problem. It is not a structurally estimated promotion rate from long-tail to established status.

The platform maximizes total discounted engagement,

$$\max_{\{x_{A,t}, x_{C,t}\}_{t \geq 0}} V(N_K; N_L) = \sum_{t=0}^{\infty} \beta^t \sum_{j \in \{A, C\}} \Theta_j \pi_j(x_{j,t}; N_{K,t}, N_L) \quad \text{s.t.} \quad N_{K,t+1} = (1 - \delta)N_{K,t} + g(Z_t, N_L). \quad (9)$$

We use engagement as the platform objective because it is directly measured in both experiments. Monetization appears as a downstream creator outcome in Study 3 but is not included in the model objective. Equivalently, the Bellman equation is

$$V(N_K; N_L) = \max_{(x_A, x_C) \in \mathcal{X}} \left\{ \sum_j \Theta_j \pi_j(x_j; N_K, N_L) + \beta V((1 - \delta)N_K + g(Z, N_L); N_L) \right\}.$$

Both pool masses affect total engagement: N_K through the baseline $v(N_K)$, the tax $c_j(N_K)$, and the future state; N_L through satiation $\gamma(N_L)$, the refreshed-pool breadth, and the development into established creators. We make the following assumptions.

ASSUMPTION 1 (Primitive domains and shape). *Let N_L lie in a compact interval $\mathcal{N}_L = [\underline{N}_L, \bar{N}_L]$ with $\underline{N}_L > 0$, and let $\mathcal{X} = [0, \bar{x}]^2$ with $0 < \bar{x} < 1$. Define $a_j = \eta_j \Theta_j$, $\bar{Z} = (a_A + a_C)\bar{x}$, and $\bar{z} = \bar{Z}/\underline{N}_L$. The primitives satisfy: $v \in C^2(\mathcal{I})$ with $v' > 0$ and $v'' < 0$; $m \in C^1(\mathcal{N}_L)$ with $m > 0$ and $m' > 0$; and $H \in C^2([0, \bar{z}])$ with $H(0) = 0$, $H' > 0$, and $H'' < 0$. The state interval is $\mathcal{I} = [0, \bar{N}]$, where $\bar{N} \geq G_{\max}/\delta$ and $G_{\max} := \max_{N_L \in \mathcal{N}_L} g(\bar{Z}, N_L)$. For every $(N_K, x, N_L) \in \mathcal{I} \times \mathcal{X} \times \mathcal{N}_L$, the current reward is jointly concave in (N_K, x) ; a sufficient primitive condition is*

$$\left(\sum_{j \in \{A, C\}} \Theta_j (1 - x_j) \right) [-v''(N_K)] \geq \frac{\Theta[v'(N_K)]^2}{\gamma(N_L)}.$$

ASSUMPTION 2 (Effective-unit routing order). *For all $N_K \in \mathcal{I}$, $q_A(N_K) := c_A(N_K)/\eta_A < q_C(N_K) := c_C(N_K)/\eta_C$. For the Study 1 blocking interpretation, the average local cost satisfies $\hat{c}(N_K) := (\Theta_A c_A(N_K) + \Theta_C c_C(N_K))/\Theta > 0$ on the relevant state range.*

ASSUMPTION 3 (No upper-cap binding for closed-form branches). *The marginal-cost formulas below are evaluated on target ranges for which the relevant segment shares are below $\bar{x}$. When a cap binds, the same KKT dispatch logic applies with projection onto $[0, \bar{x}]$.*

Assumption 1 constructs a compact invariant state space: for any $N_K \in \mathcal{I}$, $x \in \mathcal{X}$, and $N_L \in \mathcal{N}_L$, the transition is bounded by $(1 - \delta)\bar{N} + G_{\max} \leq \bar{N}$. It also imposes primitive shape restrictions. Assumption 2 states the exact mathematical routing condition. In the baseline $\eta_A = \eta_C = 1$, this reduces to the experimentally motivated cost ordering $c_A < c_C$. This ordering is setting- and intervention-specific. It should not be interpreted as implying that active viewers are universally less costly recipients of all forms of creator-support exposure. In particular, removing long tail content within the platform's usual recommendation process may generate a different viewer-segment ordering from inserting additional program-promoted content. The general allocation principle developed below applies to whichever traffic source has the lowest empirically relevant effective-unit cost. Assumption 3 is a branch condition used only to keep the closed-form cost curves transparent.

Define the exact discounted continuation return from effective attention:

$$D(Z; N_K, N_L) := \beta [V((1 - \delta)N_K + g(Z, N_L); N_L) - V((1 - \delta)N_K; N_L)]. \quad (10)$$

Appendix H.3 proves that $D(0; N_K, N_L) = 0$ and that $D(\cdot; N_K, N_L)$ is continuous, strictly increasing, and strictly concave. Let $\partial D(Z; N_K, N_L)$ denote its superdifferential with respect to Z . Any $w \in \partial D(Z; N_K, N_L)$ is a shadow value of one unit of effective attention. Only on a differentiable branch do we write $w(Z; N_K, N_L) = D'(Z; N_K, N_L)$.

The current opportunity cost of assigning long tail shares is

$$L(x_A, x_C; N_K, N_L) = \sum_{j \in \{A, C\}} \Theta_j \left(c_j(N_K) x_j + \frac{\gamma(N_L)}{2} x_j^2 \right).$$

For each feasible target $Z \in [0, \bar{Z}]$, define the minimum dispatch cost

$$C(Z; N_K, N_L) = \min_{\substack{x \in \mathcal{X} \\ a_A x_A + a_C x_C = Z}} L(x_A, x_C; N_K, N_L),$$

and let $x^d(Z; N_K, N_L)$ denote its unique minimizer. Conditional on Z , the continuation value depends only on Z , so the Bellman action problem is exactly equivalent, up to constants independent of x , to the scalar problem

$$Z^*(N_K, N_L) \in \arg \max_{Z \in [0, \bar{Z}]} \{D(Z; N_K, N_L) - C(Z; N_K, N_L)\}, \quad x^* = x^d(Z^*; N_K, N_L).$$

The objective is strictly concave because $D(\cdot; N_K, N_L)$ is strictly concave and $C(\cdot; N_K, N_L)$ is convex. Hence, the optimal target and the induced allocation are unique.

4.2. The Optimal Allocation Policy

In every period, the platform trades off the current engagement cost of long tail exposure against the future value of replenishing the established creator stock. The solution has a transparent state-contingent structure.

PROPOSITION 1 (Bellman foundation and active-first routing). *Under Assumptions 1–2, the value function $V(\cdot; N_L)$ is continuous, with a unique optimal effective-development target $Z^*(N_K, N_L)$, and a unique optimal allocation $x^*(N_K, N_L)$. There exists a shadow value $w^* \in \partial D(Z^*; N_K, N_L)$ such that, for each viewer segment $j \in \{A, C\}$,*

$$x_j^*(N_K, N_L) = \Pi_{[0, \bar{x}]} \left(\frac{\eta_j w^* - c_j(N_K)}{\gamma(N_L)} \right) = \Pi_{[0, \bar{x}]} \left(\frac{\eta_j [w^* - q_j(N_K)]}{\gamma(N_L)} \right), \quad (11)$$

where $\Pi_{[0, \bar{x}]}$ is projection onto $[0, \bar{x}]$. Viewer segments enter the long tail allocation in increasing order of effective-unit cost $q_j(N_K)$. Since $q_A(N_K) < q_C(N_K)$, active viewers enter before casual viewers. Whenever both segments are uncapped and interior,

$$x_A^*(N_K, N_L) - x_C^*(N_K, N_L) = \frac{(\eta_A - \eta_C)w^* + c_C(N_K) - c_A(N_K)}{\gamma(N_L)}.$$

This raw-share difference is positive whenever $(\eta_A - \eta_C)w^* + c_C(N_K) - c_A(N_K) > 0$, including the equal-productivity case with $c_A(N_K) < c_C(N_K)$. Moreover, for every feasible target Z and every $x \in \mathcal{X}$ satisfying $a_A x_A + a_C x_C = Z$,

$$L(x^d(Z; N_K, N_L); N_K, N_L) \leq L(x; N_K, N_L),$$

with equality if and only if $x = x^d(Z; N_K, N_L)$.

Proposition 1 characterizes the optimal routing principle for the platform. The active audience is the platform's recurrent *incubation capacity*. Because its engagement cost of long tail exposure is lowest (Study 1) and its developmental productivity may be higher (Study 3), the active viewer traffic should be routed to long tail creators with higher priority. Casual viewers should be tapped only when the development motive is strong enough to justify paying their higher effective-unit cost. For any interior solution, the KKT condition is

$$c_j(N_K) + \gamma(N_L)x_j^*(N_K, N_L) = \eta_j w^*,$$

so the platform expands long tail exposure until the current engagement cost of the marginal effective unit equals its discounted development value.

The target-and-dispatch reduction separates the quantity and routing decisions. For any chosen Z , $x^d(Z; N_K, N_L)$ minimizes current engagement cost while leaving the next state unchanged. Because the Bellman problem repeats this decision as creator capital evolves, the platform reapplies the same state-contingent routing rule each period. This is a fixed-target dispatch-efficiency result; it does not imply that segment-specific routing necessarily raises the optimal development target.

PROPOSITION 2 (Three regimes: starve, incubate, and boost). *Under Assumptions 1–3, on target ranges before feed-share caps bind, the following holds. Let $d_0(N_K, N_L) := D'_+(0; N_K, N_L)$ be the right derivative of $D(\cdot; N_K, N_L)$ at zero, and define the active-to-casual switch target*

$$Z^{\text{sw}}(N_K, N_L) = \frac{\eta_A^2 \Theta_A [q_C(N_K) - q_A(N_K)]}{\gamma(N_L)}.$$

The unique optimal target has three regimes:

- (i) **Starve:** $Z^*(N_K, N_L) = 0$ (thus, $x_A^* = x_C^* = 0$) if and only if $d_0(N_K, N_L) \leq q_A(N_K)$. In particular, if $c_A(N_K) < 0$, starvation is impossible.
- (ii) **Incubate:** $0 < Z^*(N_K, N_L) \leq Z^{\text{sw}}(N_K, N_L)$, $x_A^* = Z^*(N_K, N_L)/a_A > 0$ and $x_C^* = 0$, if and only if $d_0(N_K, N_L) > q_A(N_K)$ and $D'_+(Z^{\text{sw}}(N_K, N_L); N_K, N_L) \leq q_C(N_K)$.
- (iii) **Boost:** $Z^*(N_K, N_L) > Z^{\text{sw}}(N_K, N_L)$, $x_A^* > 0$ and $x_C^* > 0$, if and only if $d_0(N_K, N_L) > q_A(N_K)$ and $D'_+(Z^{\text{sw}}(N_K, N_L); N_K, N_L) > q_C(N_K)$.

The three regimes are state dependent. When the one-sided marginal continuation value at zero development is too low, the platform offers no developmental attention allocation to long tail creators (i.e., the **Starve** regime). When development is valuable but the target remains below the active-to-casual switch, the platform incubates its pipeline exclusively through active viewers. When the target exceeds the switch, the platform boosts development by also using casual feeds. When $c_A(N_K) < 0$, some development is a free by-product of serving active viewers the diverse content they already prefer, so the starve regime is empty. On differentiable interior branches, the regime boundaries equate the marginal return $D'(Z; N_K, N_L)$ with the marginal dispatch cost. Together, Propositions 1 and 2 characterize a recurrent policy: the platform chooses a development target in each state and fills that target from the lowest effective-unit-cost traffic source first. Active viewers emerge as the first traffic source in the baseline application of the model.

REMARK 1 (READING STUDY 1 THROUGH THE MODEL). Consider a blocking experiment that removes the same fraction d of long tail exposure from segment feeds operating at a common baseline long tail share x^u , measured in the same engagement currency as π_j . In Study 1, the realized dose is $d \approx 0.13$: blocking 50% of long tail creators' videos reduced the realized long tail view share by 13.21% (Table 2). Holding the rest of the feed fixed, per-slot measured engagement changes by

$$\Delta E_j = dx^u \left[c_j + \frac{\gamma}{2}(2-d)x^u \right], \quad (12)$$

which is positive on average by Assumption 2, strictly larger for casual than for active viewers, and approximately zero (negative if c_A is sufficiently negative) for active viewers when $c_A \approx 0$, a pattern consistent with Study 1. Under the common-baseline and common-removal assumptions, the per-exposed-slot contrast between segments, $\Delta E_C/x^u - \Delta E_A/x^u$, recovers $d(c_C - c_A)$, the cost gap that orders traffic in Propositions 1 and 2. The recovered parameters are local to the experimental ecosystem (N_K, N_L) . The experiment does not randomize segment-specific routing, but its incidence pattern motivates the active-first ordering in the model.

The model synthesizes the three empirical studies into one operating principle. Study 1 motivates heterogeneous engagement costs, Study 2 supplies a positive creator-development return, and Study 3 motivates the heterogeneous-productivity extension $\eta_A \geq \eta_C$. Propositions 1 and 2 then characterize zero, active-only, or broad long tail investment and route any positive target from lower to higher effective-unit cost. The model is partial-equilibrium and abstracts from cross-viewer interference (Johari et al. 2022), strategic creator effort, and viewer participation. Such extensions may alter the optimal development quantity, but not the conditional least-cost dispatch. This logic complements selective-exploration models (Papanastasiou et al. 2018, Che and Horner 2018): attention here is a productive input into supplier development (Wan and Wu 2017), and the platform controls feeds directly rather than through persuasion. We also remark that extensions may preserve the conditional

least-cost dispatch when they retain the separable current-cost structure and make continuation value depend on the allocation only through scalar effective attention Z . More general forms of strategic response or interference may affect both the development target and the traffic-source ranking.

5. Conclusion

This work investigates how short-video platforms should allocate viewer attention to develop long tail creators. Through two randomized field experiments and one large-scale observational study with two leading platforms in Asia, we identify the cost, return, and attention-composition factors underlying attention allocation. Study 1 identifies the engagement cost of long tail exposure and its heterogeneity across viewer segments. Study 2 establishes the causal return of additional attention for creator development, and Study 3 documents how attention composition is associated with subsequent creator outcomes. Integrating these findings, our analytical model provides a segment-aware allocation framework based on engagement cost per unit of creator-development value.

Taken together, the two experiments estimate complementary causal margins: the viewer-side consequence of changing long-tail availability and the short-run creator-side response to additional attention. Study 3 documents how subsequent creator outcomes covary with the composition of early attention. The analytical model characterizes how a platform would combine these inputs under maintained cost and productivity orderings. In the studied environments, active viewers are a plausible first routing segment, but the broader managerial implication is to estimate segment-specific effective-unit costs under the intended allocation mechanism and validate the resulting ranking through a staged routing experiment.

Practically, long tail creator development is not costless because it competes with other uses of finite viewer attention. Platforms should first estimate the current engagement cost and developmental response of each traffic source under the intended intervention, and then route developmental attention in increasing order of estimated effective-unit cost. Our framework suggests allocating attention to traffic sources with lower engagement costs per unit of creator-development value. In our setting, active viewers may serve as a lower-cost incubation channel: Study 1 shows that removing long tail content entails a smaller engagement cost for active viewers than for casual viewers, while Study 3 documents that a higher share of early exposure from active viewers is associated with stronger subsequent creator outcomes.

This work has several limitations. First, the field interventions were not sustained over extended periods because persistent traffic removal or allocation could disrupt platform operations. Longer experiments could examine whether the identified effects persist over time. Second, short-video platforms feature unusually fast feedback and algorithmic discovery, so generalizability to other content domains such as music, podcasts, or long-form video remains uncertain. Third, Study 3 is observational: temporal separation and covariate adjustment do not eliminate selection or unobserved

confounding. Randomized allocation designs are needed to identify the causal effects of attention composition. Fourth, the model is parsimonious and partial-equilibrium. It abstracts from strategic creator effort, cross-viewer interference, and platform competition. Future research could develop richer dynamic models or experimentally compare alternative attention allocation policies across viewer segments.

Endnotes

¹ TikTok; YouTube Shorts; Instagram Reels.

² This cutoff also aligns with common size-based classifications in the influencer/creator literature, where the smallest tier (“nano”) of influencers is defined as having fewer than 1,000 followers (Chen et al. 2024b).

³ The names of the partner platforms are anonymized per confidentiality agreements.

⁴ The authors have a nondisclosure agreement with Partner Platforms A and B. The data cannot be shared publicly due to the proprietary nature of the platform data and the terms of the data-sharing agreements. The empirical results reported in this paper are based on data extracts whose coverage is fully documented in Appendix B.

⁵ The user interface (UI) features of both partner platforms closely resemble TikTok’s design, as shown in Figure 1.

⁶ The content creators and viewers were unaware of this manipulation on the partner platform. The treated users accounted for less than 1% of the platform’s total traffic, ensuring minimal disruption to the overall ecosystem.

⁷ The partner platform A did not share the exact raw value of the threshold due to data sensitivity. Note that among the overall platform, active users constitute about 1.80% of the viewer population, and they account for 49.50% of total viewership and 42.09% of total video watching time. In our experiment, active viewers represent approximately 4% of the sample.

⁸ A 0.33% increase in video watching time can be operationally meaningful for a large-scale short-video platform. Applied across hundreds of millions of daily active users, even a small per-user change aggregates into a substantial shift in total platform engagement.

⁹ The HHI is calculated as follows: $HHI_c = \sum_{j=1}^M p_j^2$, where HHI_c indicates the Herfindahl-Hirschman Index for content creator c , M is the total number of unique video topics produced by creator c , and p_j represents the proportion of videos that creator c has produced on topic j , relative to their total video production.

¹⁰ Because creators and content coexist within the same platform ecosystem, interference across experimental groups cannot be completely ruled out. However, the experiment accounted for less than 1% of the platform’s total traffic, substantially limiting the scope for such interference and making any resulting spillover effects likely to be small.

¹¹ Partner Platform B allows us to observe long-term creator outcomes, which is not feasible on Partner Platform A due to a platform-wide data consolidation in late 2023 that renders creator-level performance data prior to 2024 inaccessible.

¹² For example, for a creator who registered on January 1, 2025, the initial 30-day exposure period used to construct *ActiveViewRatio* runs from January 1 to January 30. Post-exposure month 1 begins immediately after this period, and long-term outcomes are aggregated over post-exposure months 1–6. The initial exposure period is excluded from both outcome definitions.

¹³ These variables constitute the full set of creator-level covariates observable to the platform at the time of account registration, as these newly created accounts have no prior performance history or behavioral data.

References

- Abdollahpouri H, Burke R, Mobasher B (2019) Managing popularity bias in recommender systems with personalized re-ranking. *Proceedings of the Thirty-Second International Florida Artificial Intelligence Research Society Conference (FLAIRS)* (AAAI Press).
- Anderson A, Maystre L, Anderson I, Mehrotra R, Lalmas M (2020) Algorithmic effects on the diversity of consumption on spotify. *Proceedings of The Web Conference 2020 (WWW '20)*, 2155–2165 (ACM), URL <http://dx.doi.org/10.1145/3366423.3380281>.
- Bai B, Zeng Z, Zhang DJ, Xu Z (2026) The impact of generative AI search on content-sharing platforms, URL <http://dx.doi.org/10.2139/ssrn.6171026>, sSRN Working Paper No. 6171026; major revision at Manufacturing & Service Operations Management.
- Besbes O, Gur Y, Zeevi A (2016) Optimization in online content recommendation services: Beyond click-through rates. *Manufacturing & Service Operations Management* 18(1):15–33, URL <http://dx.doi.org/10.1287/msom.2015.0548>.
- Bhargava HK (2022) The creator economy: Managing ecosystem supply, revenue sharing, and platform design. *Management Science* 68(7):5233–5251, URL <http://dx.doi.org/10.1287/mnsc.2021.4126>.
- Brown MA, Bisbee J, Lai A, Bonneau R, Nagler J, Tucker JA (2022) Echo chambers, rabbit holes, and algorithmic bias: How YouTube recommends content to real users, URL <https://ssrn.com/abstract=4114905>, sSRN.
- Burtch G, He Q, Hong Y, Lee D (2022) How do peer awards motivate creative content? Experimental evidence from Reddit. *Management Science* 68(5):3488–3506, URL <http://dx.doi.org/10.1287/mnsc.2021.4040>.
- Cao Z, Zhu Y, Li G, Qiu L (2023) Consequences of information feed integration on user engagement and contribution: A natural experiment in an online knowledge-sharing community. *Information Systems Research* 35(3):1114–1136, URL <http://dx.doi.org/10.1287/isre.2022.0043>.
- Carroni E, Madio L, Shekhar S (2024) Superstar exclusivity in two-sided markets. *Management Science* 70(2):991–1011, URL <http://dx.doi.org/10.1287/mnsc.2023.4720>.
- Chaudhary T (2025) Short video platform market size, share, report, forecast 2035. Market research report, Market Research Future.
- Che YK, Horner J (2018) Recommender systems as mechanisms for social learning. *Quarterly Journal of Economics* 133(2):871–925, URL <http://dx.doi.org/10.1093/qje/qjx044>.
- Chen G, Chan T, Zhang D, Liu S, Wu Y (2024a) The impact of a more diversified recommender system on digital platforms: Evidence from a large-scale field experiment, URL <http://dx.doi.org/10.2139/ssrn.4365121>, sSRN Working Paper No. 4365121.

-
- Chen J, Zhang Y, Cai H, Liu L, Liao M, Fang J (2024b) A comprehensive overview of micro-influencer marketing. *Behavioral Sciences* 14(3):243, URL <http://dx.doi.org/10.3390/bs14030243>.
- Cheng Z, Nault BR (2012) Relative industry concentration and customer-driven IT spillovers. *Information Systems Research* 23(2):340–355, URL <http://dx.doi.org/10.1287/isre.1100.0345>.
- Claussen J, Peukert C, Sen A (2019) The editor vs. the algorithm: Returns to data and externalities in online news. CESifo Working Paper No. 8012, CESifo.
- Covington P, Adams J, Sargin E (2016) Deep neural networks for YouTube recommendations. *Proceedings of the 10th ACM Conference on Recommender Systems (RecSys '16)*, 191–198 (ACM), URL <http://dx.doi.org/10.1145/2959100.2959190>.
- Cui R, Li J, Zhang DJ (2020) Reducing discrimination with reviews in the sharing economy. *Management Science* 66(3):1071–1094, URL <http://dx.doi.org/10.1287/mnsc.2018.3273>.
- Cutolo D, Kenney M (2021) Platform-dependent entrepreneurs: Power asymmetries, risks, and strategies. *Academy of Management Perspectives* 35(4):584–605, URL <http://dx.doi.org/10.5465/amp.2019.0103>.
- Duan W, Gu B, Whinston AB (2008) Do online reviews matter? An empirical investigation of panel data. *Decision Support Systems* 45(4):1007–1016, URL <http://dx.doi.org/10.1016/j.dss.2008.04.001>.
- Forderer J, Gutt D, Greenwood BN (2025) Star wars: An empirical study of star performer turnover and content supply on multisided streaming platforms. *Information Systems Research* 37(1):568–583, URL <http://dx.doi.org/10.1287/isre.2023.0367>.
- Gaenssle S, Budzinski O (2021) Stars in social media: New light through old windows? *Journal of Media Business Studies* 18(2):79–105, URL <http://dx.doi.org/10.1080/16522354.2020.1738694>.
- Gallus J (2017) Fostering public good contributions with symbolic awards: A large-scale natural field experiment at Wikipedia. *Management Science* 63(12):3999–4015, URL <http://dx.doi.org/10.1287/mnsc.2016.2540>.
- Goes PB, Guo C, Lin M (2016) Do incentive hierarchies induce user effort? Evidence from an online knowledge exchange. *Information Systems Research* 27(3):497–516, URL <http://dx.doi.org/10.1287/isre.2016.0635>.
- Greenstein S, Gu G, Zhu F (2021) Ideology and composition among an online crowd: Evidence from wikipedians. *Management Science* 67(5):3067–3086, URL <http://dx.doi.org/10.1287/mnsc.2020.3661>.
- Gu M, Liu D, Kumar S (2024) Navigating platform-led affiliate marketing: Implications for content creation and platform profitability. *Information Systems Research* 36(2):802–827, URL <http://dx.doi.org/10.1287/isre.2022.0620>.
- Gupta V, Han BR, Kim SH, Paek H (2020) Maximizing intervention effectiveness. *Management Science* 66(12):5576–5598, URL <http://dx.doi.org/10.1287/mnsc.2019.3537>.
- Han S, Ravid SA (2020) Star turnover and the value of human capital: Evidence from Broadway shows. *Management Science* 66(2):958–978, URL <http://dx.doi.org/10.1287/mnsc.2018.3177>.
- Hayes AF (2022) *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach* (Guilford Press), 3rd edition.
- Heinonen K (2011) Consumer activity in social media: Managerial approaches to consumers’ social media behavior. *Journal of Consumer Behaviour* 10(6):356–364, URL <http://dx.doi.org/10.1002/cb.376>.

-
- Hu M, Benjaafar S (2020) Operations management in the age of the sharing economy: What is old and what is new? *Manufacturing & Service Operations Management* 22(1):93–101, URL <http://dx.doi.org/10.1287/msom.2019.0803>.
- Hu Q, Huang N, Zhang R (2024) Traffic allocation optimization on video sharing platforms: Balancing viewer engagement and small creator development. *Proceedings of the 45th International Conference on Information Systems* (Bangkok, Thailand), URL https://aisel.aisnet.org/icis2024/sharing_econ/sharing_econ/4.
- Huang N, Burtch G, Gu B, Hong Y, Liang C, Wang K, Fu D, Yang B (2019) Motivating user-generated content with performance feedback: Evidence from randomized field experiments. *Management Science* 65(1):327–345, URL <http://dx.doi.org/10.1287/mnsc.2017.2944>.
- Jain S, Qian K (2021) Compensating online content producers: A theoretical analysis. *Management Science* 67(11):7075–7090, URL <http://dx.doi.org/10.1287/mnsc.2020.3862>.
- Johari R, Li H, Liskovich I, Weintraub GY (2022) Experimental design in two-sided platforms: An analysis of bias. *Management Science* 68(10):7069–7089, URL <http://dx.doi.org/10.1287/mnsc.2021.4247>.
- Kellogg KC, Valentine MA, Christin A (2020) Algorithms at work: The new contested terrain of control. *Academy of Management Annals* 14(1):366–410, URL <http://dx.doi.org/10.5465/annals.2018.0174>.
- Kim Y, Kim K, Park C, Yu H (2019) Sequential and diverse recommendation with long tail. *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI 2019)*, 2740–2746 (ijcai.org), URL <http://dx.doi.org/10.24963/ijcai.2019/380>.
- Knight B, Mitrofanov D, Netessine S (2026) Human–algorithm collaboration in gig work: The role of experience, skill level, and task complexity. *Information Systems Research* URL <http://dx.doi.org/10.1287/isre.2024.1664>, articles in Advance.
- Kumar A, Smith MD, Telang R (2014) Information discovery and the long tail of motion picture content. *MIS Quarterly* 38(4):1057–1078, URL <http://dx.doi.org/10.25300/MISQ/2014/38.4.06>.
- Lambrecht A, Tucker C (2019) Algorithmic bias? An empirical study of apparent gender-based discrimination. *Management Science* 65(7):2966–2981, URL <http://dx.doi.org/10.1287/mnsc.2018.3093>.
- Levina N, Arriaga M (2014) Distinction and status production on user-generated content platforms: Using bourdieu’s theory of cultural production to understand social dynamics in online fields. *Information Systems Research* 25(3):468–488, URL <http://dx.doi.org/10.1287/isre.2014.0535>.
- Li J, Netessine S (2020) Higher market thickness reduces matching rate in online platforms: Evidence from a quasi-experiment. *Management Science* 66(1):271–289, URL <http://dx.doi.org/10.1287/mnsc.2018.3223>.
- Liang C, Hong Y, Gu B (2024) Monitoring and the cold start problem in digital platforms: Theory and evidence from online labor markets. *Information Systems Research* 36(1):600–620, URL <http://dx.doi.org/10.1287/isre.2021.0146>.
- Lika B, Kolomvatsos K, Hadjiefthymiades S (2014) Facing the cold start problem in recommender systems. *Expert Systems with Applications* 41(4):2065–2073, URL <http://dx.doi.org/10.1016/j.eswa.2013.09.005>.
- Liu W, Xi Y, Qin J, Sun F, Chen B, Zhang W, Zhang R, Tang R (2022) Neural re-ranking in multi-stage recommender systems: A review. Preprint arXiv:2202.06602, arXiv, arXiv:2202.06602.

- Long X, Sun J, Dai H, Zhang D, Zhang J, Chen Y, Hu H, Zhao B (2024) The choice overload effect in online recommender systems. *Manufacturing & Service Operations Management* 27(1):249–268, URL <http://dx.doi.org/10.1287/msom.2022.0659>.
- Maclean KDS (2021) Value of stars on Broadway: A case study. *Service Science* 13(2):77–87, URL <http://dx.doi.org/10.1287/serv.2021.0273>.
- Oestreicher-Singer G, Zalmanson L (2013) Content or community? A digital business strategy for content providers in the social age. *MIS Quarterly* 37(2):591–616, URL <http://dx.doi.org/10.25300/MISQ/2013/37.2.12>.
- Papanastasiou Y, Bimpikis K, Savva N (2018) Crowdsourcing exploration. *Management Science* 64(4):1727–1746, URL <http://dx.doi.org/10.1287/mnsc.2016.2697>.
- Qian K, Jain S (2024) Digital content creation: An analysis of the impact of recommendation systems. *Management Science* 70(12):8668–8684, URL <http://dx.doi.org/10.1287/mnsc.2022.03655>.
- Qian K, Xie Y (2026) The power of star creators: Evidence from the live streaming industry. *Journal of Marketing Research* 63(1):149–166, URL <http://dx.doi.org/10.1177/00222437251359163>.
- Rafeian O (2023) Optimizing user engagement through adaptive content recommendation. *Management Science* 69(12):7854–7872, URL <http://dx.doi.org/10.1287/mnsc.2023.4710>.
- Ren Q (2024) Navigating the creator economy: An analysis of content promotion and view allocation policies on digital content platforms. *Marketing Science* 44(3):671–690, URL <http://dx.doi.org/10.1287/mksc.2023.0573>.
- Stuart EA (2010) Matching methods for causal inference: A review and a look forward. *Statistical Science* 25(1):1–21.
- Sun T, Yuan Z, Li C, Zhang K, Xu J (2024) The value of personal data in internet commerce: A high-stakes field experiment on data regulation policy. *Management Science* 70(4):2645–2660, URL <http://dx.doi.org/10.1287/mnsc.2023.4828>.
- Wan Z, Wu B (2017) When suppliers climb the value chain: A theory of value distribution in vertical relationships. *Management Science* 63(2):477–496.
- Wu B, Liu CZ, Choo KKR, Tsuyuguchi T (2023) Voters’ impacts on creators’ popularity disparity and network size. *Proceedings of the 56th Hawaii International Conference on System Sciences (HICSS 2023)* (ScholarSpace).
- Wu Y, Zhu F (2022) Competition, contracts, and creativity: Evidence from novel writing in a platform market. *Management Science* 68(12):8613–8634, URL <http://dx.doi.org/10.1287/mnsc.2022.4329>.
- Xia T (2025) Supporting content creators on two-sided markets: Experimental evidence from a short-form video platform, URL <http://dx.doi.org/10.2139/ssrn.5118719>, SSRN Working Paper No. 5118719, first posted February 19, 2025.
- Xu Y, Dai H, Yan W (2024) Identity disclosure and anthropomorphism in voice chatbot design: A field experiment. *Management Science* 72(1):223–241, URL <http://dx.doi.org/10.1287/mnsc.2022.03833>.
- Xu Y, Lu B, Ghose A, Dai H, Zhou W (2023) The interplay of earnings, ratings, and penalties on sharing platforms: An empirical investigation. *Management Science* 69(10):6128–6146, URL <http://dx.doi.org/10.1287/mnsc.2023.4761>.
- Yin H, Cui B, Li J, Yao J, Chen C (2012) Challenging the long tail recommendation. *Proceedings of the VLDB Endowment* 5(9):896–907, URL <http://dx.doi.org/10.14778/2311906.2311916>.

- Zeng Z, Dai H, Zhang DJ, Zhang H, Zhang R, Xu Z, Shen ZJM (2023) The impact of social nudges on user-generated content for social network platforms. *Management Science* 69(9):5189–5208, URL <http://dx.doi.org/10.1287/mnsc.2022.4622>.
- Zeng Z, Zhang Z, Zhang D, Chan T (2024) The impact of recommender systems on content consumption and production: Evidence from field experiments and structural modeling, URL <http://dx.doi.org/10.2139/ssrn.4915562>, SSRN Working Paper No. 4915562.
- Zhang J, Moon K, Veeraraghavan SK (2022) Does fake news create echo chambers?, URL <http://dx.doi.org/10.2139/ssrn.4144897>, SSRN Working Paper No. 4144897.
- Zhang K, Wang Q, Qiu L, Wang N (2024) Unveiling the cost of free: How an ad-sponsored model affects serialized digital content creation. *Information Systems Research* 36(2):962–982, URL <http://dx.doi.org/10.1287/isre.2023.0263>.
- Zhang P, Cui R, Zhang DJ (2026) The impact of AI search on the online content ecosystem: Evidence from Google and Reddit, URL <http://dx.doi.org/10.2139/ssrn.6762279>, SSRN Working Paper No. 6762279.
- Zhang X, Zhu F (2011) Group size and incentives to contribute: A natural experiment at Chinese Wikipedia. *American Economic Review* 101(4):1601–1615, URL <http://dx.doi.org/10.1257/aer.101.4.1601>.
- Zhao K, Hong Y, Ma T, Lu Y, Hu Y (2025) Lost in the crowd: How group size and content moderation shape user engagement in live streaming. *Information Systems Research* 36(4):2076–2095, URL <http://dx.doi.org/10.1287/isre.2022.0086>.
- Zhou R, Khemmarat S, Gao L (2010) The impact of YouTube recommendation system on video views. *Proceedings of the 10th ACM SIGCOMM Conference on Internet Measurement (IMC '10)*, 404–410 (ACM), URL <http://dx.doi.org/10.1145/1879141.1879193>.
- Zhu F, Zhang X (2010) Impact of online consumer reviews on sales: The moderating role of product and consumer characteristics. *Journal of Marketing* 74(2):133–148, URL <http://dx.doi.org/10.1509/jm.74.2.133>.
- Zou T, Wu Y, Sarvary M (2025) Quality-variety tradeoffs in recommendation systems on content platforms. *Management Science* URL <http://dx.doi.org/10.1287/mnsc.2025.00300>.

Online Appendices

Attention as Investment: Traffic Allocation for Long Tail Creators

Appendix A: Recommender Systems

The partner platforms utilize state-of-the-art deep learning algorithms in their recommender systems, aligning with industry standards, to deliver video recommendations to viewers. As illustrated in Figure A1, a typical recommender system on content sharing platforms operates through three sequential phases: retrieval, ranking, and re-ranking (Covington et al. 2016, Liu et al. 2022). In the retrieval phase, the system selects a broad set of videos (500–10,000) from the video pool for each viewer by computing the proximity between user and video embeddings. In the ranking phase, the algorithm ranks these videos by their anticipated match value using deep neural networks designed to predict user behaviors, shortlisting the top 100–300. In the re-ranking phase, the system re-ranks the shortlisted videos to determine the final 10–20 recommendations, which are sorted by viewer preference and presented on the platform’s user interface. If the viewer continues to engage after consuming these recommendations, the system dynamically excludes previously recommended videos and repeats the three phases to refresh the feed.

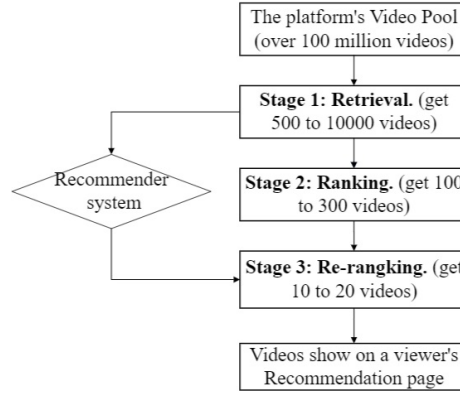


Figure A1: Workflow of a Typical Recommender System on Content Sharing Platforms

Appendix B: Data Description and Summary Statistics

Tables B1 and B2 present variable definitions and descriptive statistics for Study 1 (viewer-side experiment). Tables B3 and B4 present variable definitions and descriptive statistics for Study 2 (creator-side experiment). Tables B5–B6 present variable definitions, and Table B7 presents descriptive statistics for Study 3 (observational analysis of newly registered creators).

The data available for each study are as follows. Study 1 data are at the viewer level ($N = 13,597,545$) and include demographics (gender, age, city level, account tenure), engagement metrics (watching time, likes, comments, forwards, complete views), and content diversity measures (topics browsed, HHI, entropy), each observed during the week before and the 10-day experiment period. A separate cross-sectional dataset of 188,374,718 creators from Platform A, covering a 7-day window, was used for the mechanism analysis (content quality and content diversity comparisons). Study 2 data are at the creator level ($N = 960,401$) and include demographics, viewer engagement with the creator’s content (views, likes, comments, forwards), production quantity (videos posted and total video length), content quality (completion rate, fast-swipe rate, engagement rates), growth (net new followers), effort (production days), and the creator’s own consumption behavior as a viewer on the platform, each measured during the week before and the 15-day experiment period. Study 3 data are at the creator level ($N = 885,478$) and include registration demographics (gender, city level, region, community type), active-viewer composition measured during an initial 30-day exposure period, short-term outcomes measured during post-exposure month 1, long-term outcomes aggregated over post-exposure months 1–6, and monthly panel data tracking productivity, performance, and growth. To protect data confidentiality across all three studies, each partner platform multiplied continuous variables by a random, undisclosed constant before sharing the data; all numerical values therefore represent scaled metrics, and coefficients should be interpreted in relative terms.

Table B1: Variable Descriptions in Study 1

Variable	Description
Independent Variable	
Treatment	A viewer’s assignment to the experimental group (0 = Control, 1 = Treatment)
Viewer Demographics	
Gender	A gender indicator (0 = Unknown, 1 = Male, 2 = Female)
CityLevel	The tier of a viewer’s located city (0 = Unknown, 1 = First-tier city, 2 = Second-tier city, 3 = Third-tier city, 4 = Fourth-tier city, 5 = Fifth-tier and over Fifth-tier city)
NumFriends	The total number of friends that a viewer has on the platform
NumFollowers	The total number of followers that a viewer has on the platform
Viewer Engagement	
WatchingTime	The total duration of video watched by a viewer during the experiment (in seconds)
NumComments	The total number of comments a viewer posted on videos during the experiment
NumForwards	The total number of videos a viewer forwarded to others during the experiment
NumLikes	The total number of likes a viewer left on videos during the experiment
NumCompleteViews	The number of video views in which the viewer watched at least 90% of the video’s total duration during the experiment period
Others	
Active	A binary indicator of whether a viewer is classified as active by the platform (0 = Casual, 1 = Active)
NumViews	The total number of videos presented to the viewer in their feed during the experiment, regardless of the watching duration
LongTailViewShare	The proportion of video views that come from long tail creator content relative to the total video views by a viewer during the experiment

Notes: To protect data confidentiality, the partner platform multiplied each variable by a random, undisclosed constant. All variables are measured at the viewer level.

Table B2: Descriptive Statistics in Study 1

Variable	Obs.	Mean	Std	Min	Max
Viewer Demographics					
<i>Gender</i>	13 597 545	1.48	0.52	0	2
<i>CityLevel</i>	13 597 545	3.40	1.29	0	5
<i>NumFriends</i>	13 597 545	456.80	704.23	0	22 288
<i>NumFollowers</i>	13 597 545	0.40	6.24	0	4787
Viewer Engagement					
<i>WatchingTime</i>	13 597 545	7659.90	19 364.27	0	637 114
<i>NumComments</i>	13 597 545	0.22	3.45	0	550
<i>NumForwards</i>	13 597 545	1.00	6.95	0	550
<i>NumLikes</i>	13 597 545	2.24	13.47	0	550
<i>NumCompleteViews</i>	13 597 545	134.45	330.91	0	14 853
Others					
<i>Active</i>	13 597 545	0.04	0.20	0	1
<i>NumViews</i>	13 597 545	333.60	901.29	0	61 923
<i>LongTailViewShare</i>	13 597 545	0.10	0.12	0	1

Notes: (a) To protect data confidentiality, the partner platform scaled the raw values of several variables (NumFriends, NumFollowers, WatchingTime, NumComments, NumForwards, NumLikes, NumCompleteViews, NumViews). Therefore, the numerical values presented in the table represent scaled metrics rather than absolute values. (b) All variables are measured at the viewer level. (c) NumViews and LongTailViewShare are used in the Manipulation Checks in the main text. (d) NumComments, NumForwards, and NumLikes are capped at platform-defined upper thresholds to exclude outliers and abnormal activity.

Table B3: Variable Descriptions in Study 2

Variable	Description
Independent Variable	
Treatment	A creator’s assignment to the experimental group (0 = Control, 1 = Treatment)
Creator Demographics	
Gender	A gender indicator (0 = Unknown, 1 = Male, 2 = Female)
CityLevel	The tier of a creator’s located city (0 = Unknown, 1 = First-tier city, 2 = Second-tier city, 3 = Third-tier city, 4 = Fourth-tier city, 5 = Fifth-tier and over Fifth-tier city)
NumFollowers	The total number of followers that a creator has on the platform
Creator Productivity	
NumUploadVideos	The total number of videos uploaded by a creator during the experiment
ProducedVideoLength	The total length of the videos produced by a creator during the experiment (in seconds)
DaysOfProduction	The total number of days a creator produced videos during the experiment
Creator Performance	
NumLikes	The total number of likes on a creator’s video content during the experiment
NumForwards	The total number of forwards on a creator’s video content during the experiment
NumComments	The total number of comments on a creator’s video content during the experiment
NumCompleteViews	The number of video views on a creator’s content in which viewers watched at least 90% of a video’s total duration during the experiment
Creator Growth	
NumNewFollowers	The total number of new followers gained by a creator during the experiment
MonetizeProbability	A binary indicator for whether a creator became eligible for platform monetization during the experiment, based on the platform’s criteria (0 = Ineligible, 1 = Eligible)
Others	
NumViews	The total number of times the creator’s content is presented to viewers in their feeds

Notes: To protect data confidentiality, the partner platform multiplied each variable by a random, undisclosed constant. All variables are measured at the creator level.

Table B4: Descriptive Statistics in Study 2

Variable	Obs.	Mean	Std	Min	Max
Creator Demographics					
<i>Gender</i>	960 401	1.63	0.48	0	2
<i>CityLevel</i>	960 401	3.44	1.39	0	5
<i>NumFollowers</i>	960 401	10.31	44.92	0	999
Creator Productivity					
<i>NumUploadVideos</i>	960 401	3.06	17.88	0	4690
<i>ProducedVideoLength</i>	960 401	1044.11	8490.49	0	3 297 589
<i>DaysOfProduction</i>	960 401	2.11	6.44	0	73
Creator Performance					
<i>NumLikes</i>	960 401	109.29	27 640.15	0	22 095 603
<i>NumForwards</i>	960 401	34.33	7674.89	0	6 080 395
<i>NumComments</i>	960 401	3.68	545.02	0	340 697
<i>NumCompleteViews</i>	960 401	1986.29	332 579.93	0	151 806 470
Creator Growth					
<i>NumNewFollowers</i>	960 401	9.58	1571.53	−1795	658 610
<i>MonetizeProbability</i>	960 401	0.02	0.12	0	1
Others					
<i>NumViews</i>	960 401	10 277.46	1 087 184.36	0	541 603 041

Notes: (a) To protect data confidentiality, the partner platform scaled the raw values of several variables (NumUploadVideos, ProducedVideoLength, DaysOfProduction, NumLikes, NumForwards, NumComments, NumCompleteViews, NumNewFollowers, NumViews). Therefore, the numerical values presented represent scaled metrics. (b) All variables are measured at the creator level.

Table B5: Variable Descriptions in Study 3 (Part I — Independent Variable and Short-Term Measures)

Variable	Description
Independent Variable	
ActiveViewRatio	The proportion of views from active viewers relative to total views during each creator's initial 30-day exposure period following registration
Short-Term Creator Productivity	
NumUploadVideos	The total number of videos uploaded by a creator during post-exposure month 1
ProducedVideoLength	The total length of the videos produced by a creator during post-exposure month 1 (in seconds)
DaysOfProduction	The total number of days a creator produced videos during post-exposure month 1
Short-Term Creator Performance	
NumLikes	The total number of likes on a creator's video content during post-exposure month 1
NumForwards	The total number of forwards on a creator's video content during post-exposure month 1
NumComments	The total number of comments on a creator's video content during post-exposure month 1
NumCompleteViews	The number of video views on a creator's video content in which the viewer watched at least 90% of the video's total duration during post-exposure month 1
Short-Term Creator Growth	
NumNewFollowers	The total number of new followers gained by a creator during post-exposure month 1
MonetizeProbability	A binary indicator for whether a creator became eligible for platform monetization during post-exposure month 1, based on the platform's criteria (0 = Ineligible, 1 = Eligible)

Notes: (a) We sample creators who registered between January 1, 2025, and January 14, 2025. (b) *ActiveViewRatio* is measured during the initial 30-day exposure period following registration. (c) Short-term dependent variables are measured during post-exposure month 1. The exposure and outcome windows do not overlap.

Table B6: Variable Descriptions in Study 3 (Part II — Long-Term Measures and Control Variables)

Variable	Description
Long-Term Creator Productivity	
NumUploadVideos	The total number of videos uploaded by a creator over post-exposure months 1–6
ProducedVideoLength	The total length of the videos produced by a creator over post-exposure months 1–6 (in seconds)
DaysOfProduction	The total number of days a creator produced videos over post-exposure months 1–6
Long-Term Creator Performance	
NumLikes	The total number of likes on a creator's video content over post-exposure months 1–6
NumForwards	The total number of forwards on a creator's video content over post-exposure months 1–6
NumComments	The total number of comments on a creator's video content over post-exposure months 1–6
NumCompleteViews	The number of video views on a creator's video content in which the viewer watched at least 90% of the video's total duration over post-exposure months 1–6
Long-Term Creator Growth	
NumNewFollowers	The total number of new followers gained by a creator over post-exposure months 1–6
MonetizeProbability	A binary indicator for whether a creator became eligible for platform monetization over post-exposure months 1–6, based on the platform's criteria (0 = Ineligible, 1 = Eligible)
Control Variables	
Gender	A gender indicator (0 = Unknown, 1 = Male, 2 = Female)
CityLevel	The tier of a creator's located city (0 = Unknown, 1 = First-tier city, 2 = Second-tier city, 3 = Third-tier city, 4 = Fourth-tier city, 5 = Fifth-tier and over Fifth-tier city)
Region	Region of the creator (0 = Unknown; 1 = Northern; 2 = Southern)
CommunityType	Community classification of the creator (0 = Unknown; 1 = Urban; 2 = Foreign; 3 = Rural; 4 = Township)

Notes: (a) Long-term dependent variables are aggregated over post-exposure months 1–6. The initial 30-day exposure period is excluded from all dependent variables. (b) Control variables are measured at the time of creator registration.

Table B7: Descriptive Statistics in Study 3

Variable	Obs.	Mean	Std	Min	Max
Independent Variable					
<i>ActiveViewRatio</i>	885 478	0.83	0.22	0	1
Short-Term Creator Productivity					
<i>NumUploadVideos</i>	885 478	6.91	99.53	0	6436.25
<i>ProducedVideoLength</i>	885 478	325.15	6897.34	0	799 724.27
<i>DaysOfProduction</i>	885 478	1.76	4.76	0	37.50
Short-Term Creator Performance					
<i>NumLikes</i>	885 478	37.59	265.95	0	26 716.25
<i>NumForwards</i>	885 478	5.06	102.91	0	57 772.50
<i>NumComments</i>	885 478	10.70	91.54	0	22 756.25
<i>NumCompleteViews</i>	885 478	48.55	536.10	0	77 478.75
Short-Term Creator Growth					
<i>NumNewFollowers</i>	885 478	3.11	183.48	−8523.75	45 056.25
<i>MonetizeProbability</i>	885 478	0.00	0.02	0	1
Long-Term Creator Productivity					
<i>NumUploadVideos</i>	885 478	21.22	258.96	0	18 647.50
<i>ProducedVideoLength</i>	885 478	853.61	15 753.29	0	2 291 924.35
<i>DaysOfProduction</i>	885 478	6.99	19.81	0	225.00
Long-Term Creator Performance					
<i>NumLikes</i>	885 478	195.84	1070.70	0	53 917.50
<i>NumForwards</i>	885 478	27.19	370.35	0	77 458.75
<i>NumComments</i>	885 478	56.25	398.44	0	83 402.50
<i>NumCompleteViews</i>	885 478	188.13	1197.87	0	97 200.00
Long-Term Creator Growth					
<i>NumNewFollowers</i>	885 478	12.97	375.00	−20 491.25	37 363.75
<i>MonetizeProbability</i>	885 478	0.00	0.03	0	1
Control Variables					
<i>Gender</i>	885 478	1.29	0.62	0	2
<i>CityLevel</i>	885 478	2.95	1.53	0	5
<i>Region</i>	885 478	1.60	0.61	0	2
<i>CommunityType</i>	885 478	1.08	1.45	0	4

Notes: (a) *ActiveViewRatio* is measured during the initial 30-day exposure period following registration; short-term outcomes are measured during post-exposure month 1; long-term outcomes are aggregated over post-exposure months 1–6. The exposure and outcome windows do not overlap. (b) To protect data confidentiality, the partner platform scaled the raw values of several variables. (c) All variables are measured at the creator level.

Appendix C: Validation Checks

Tables C1 and C2 report the randomization checks for Studies 1 and 2, confirming that treatment and control groups are balanced on all pre-experiment characteristics (all $p > 0.05$). Table C3 reports balance checks for Study 3, comparing newly registered creators above and below the median *ActiveViewRatio* on all observable demographic and geographic characteristics at registration. The absolute standardized mean differences range from 0.002 to 0.225. Although none exceeds 0.25, some exceed the more stringent 0.10 benchmark; we therefore interpret these checks as evidence against severe observable imbalance rather than as evidence of complete covariate balance (Stuart 2010).

Table C1: Randomization Checks in Study 1

Variable	Treatment Mean	Control Mean	<i>p</i> -value
Viewer Demographics			
Gender	1.48	1.48	0.11
CityLevel	3.40	3.40	0.68
NumFriends	456.42	456.99	0.16
NumFollowers	0.40	0.40	0.32
Viewer Engagement			
WatchingTime	5417.88	5404.59	0.10
NumComments	0.15	0.15	0.91
NumForwards	0.65	0.65	0.41
NumLikes	1.49	1.49	0.51
NumCompleteViews	94.90	95.13	0.67
Others			
NumViews	229.28	228.81	0.19
LongTailViewShare	0.11	0.11	0.57

Notes: (a) To protect data confidentiality, the partner platform multiplied each variable by a random, undisclosed constant. Therefore, the numerical values represent scaled metrics rather than absolute values. (b) All variables are measured at the viewer level. (c) For randomization checks, we apply two-tailed t-tests to continuous variables and chi-square tests to categorical variables.

Table C2: Randomization Checks in Study 2

Variable	Treatment Mean	Control Mean	<i>p</i> -value
Creator Demographics			
Gender	1.63	1.63	0.63
CityLevel	3.44	3.44	0.63
NumFollowers	10.35	10.26	0.32
Creator Productivity			
NumUploadVideos	3.02	2.98	0.19
ProducedVideoLength	1015.77	996.46	0.19
DaysOfProduction	2.13	2.11	0.14
Creator Performance			
NumLikes	9.79	10.65	0.55
NumForwards	5.16	4.72	0.39
NumComments	0.56	0.53	0.62
NumCompleteViews	148.19	318.25	0.06
Creator Growth			
NumNewFollowers	2.27	2.24	0.77
MonetizeProbability	0.02	0.02	0.95
Others			
NumViews	2428.36	2864.50	0.13

Notes: (a) To protect data confidentiality, the partner platform multiplied each variable by a random, undisclosed constant. Therefore, the numerical values represent scaled metrics rather than absolute values. (b) For randomization checks, we apply two-tailed t-tests to continuous variables and chi-square tests to categorical variables.

Table C3: Balance Checks in Study 3

Variable	Level	SMD	Variable	Level	SMD
Gender			CommunityType		
Gender	0	0.225	CommunityType	0	0.033
Gender	1	0.014	CommunityType	1	0.048
Gender	2	0.146	CommunityType	2	0.006
CityLevel			CommunityType	3	0.077
CityLevel	0	0.027	CommunityType	4	0.010
CityLevel	1	0.057	Region		
CityLevel	2	0.078	Region	0	0.027
CityLevel	3	0.002	Region	1	0.125
CityLevel	4	0.073	Region	2	0.131
CityLevel	5	0.059			

Notes: (a) Level indicates the category value of each categorical variable (e.g., Gender = 0–2; CityLevel = 0–5; CommunityType = 0–4; Region = 0–2). (b) |SMD| represents the absolute standardized mean difference between groups above and below the median *ActiveViewRatio*. All SMD values are below the commonly used threshold of 0.25 (Stuart, 2010).

Appendix D: Additional Content-Quality Checks in Study 2

Study 2 shows that allocating additional viewer attention to long tail creators increases their content production, performance, and follower growth. A potential concern is that the increase in production may come at the expense of content quality. To examine this possibility, we re-estimate Equation (3) using four engagement-rate measures as proxies for content quality: like rate, comment rate, forward rate, and complete-view rate. Because these measures are defined only for creators who produced content during the experiment, the analysis is restricted to the 182,753 creators who uploaded at least one video during the experiment period.

Table D4 reports the results. Conditional on producing at least one video during the experiment, the treatment coefficients for like rate, comment rate, and complete-view rate are statistically insignificant. The treatment is positively and marginally significantly associated with forward rate, corresponding to a 3.36% increase relative to the control-group mean ($p < 0.10$). Thus, among creators who posted during the experiment, a post-treatment selected sample, the observed engagement-rate measures do not decline. We treat this as a descriptive conditional check rather than a causal estimate of the treatment effect on content quality. Overall, we find no evidence that the increase in content production induced by additional viewer attention is accompanied by deterioration in these observed content quality measures.

Table D4: Content Quality Results in Study 2

Variable	<i>LikeRate</i> (1)	<i>CommentRate</i> (2)	<i>ForwardRate</i> (3)	<i>CompleteViewRate</i> (4)
<i>Treatment</i>	0.001 (0.000)	−0.0003 (0.001)	0.001* (0.000)	−0.0001 (0.001)
<i>Relative Effect Size (%)</i>	0.81	−0.37	3.36*	−0.29
Observations	182,753	182,753	182,753	182,753

Notes: This table reports conditional OLS estimates based on creators who uploaded at least one video during the experiment period. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Appendix E: Specification Checks in Study 3

Table E1, Panels A–C and Table E2, Panels A–C present short- and long-term results from models excluding all control variables. Table E3, Panels A–C and Table E4, Panels A–C present the corresponding results using log-transformed outcomes. The log-transformed specifications and most no-control estimates point in the same direction as the controlled main results. Two long-term no-control estimates are less stable: the coefficient on new followers becomes statistically insignificant, and the coefficient on forwards becomes small, negative, and statistically insignificant. We therefore treat the controlled specifications reported in the main text as the primary estimates and the no-control models as sensitivity checks rather than as evidence that every result is invariant to covariate adjustment.

Table E1: Short-Term Results Without Controls in Study 3

Panel A: Creator Productivity

Variable	(1)	(2)	(3)
	<i>NumUploadVideos</i>	<i>ProducedVideoLength</i>	<i>DaysOfProduction</i>
<i>ActiveViewRatio</i>	6.993*** (0.340)	355.442*** (24.357)	0.916*** (0.016)
<i>Controls</i>	No	No	No
Observations	885 478	885 478	885 478

Panel B: Creator Performance

Variable	(1)	(2)	(3)	(4)
	<i>NumCompleteViews</i>	<i>NumLikes</i>	<i>NumComments</i>	<i>NumForwards</i>
<i>ActiveViewRatio</i>	35.033*** (1.552)	11.592*** (0.603)	3.532*** (0.236)	0.725*** (0.221)
<i>Controls</i>	No	No	No	No
Observations	885 478	885 478	885 478	885 478

Panel C: Creator Growth

Variable	(1)	(2)	(3)	(4)
	<i>NumNewFollowers</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>	<i>MonetizeProbability</i>
<i>ActiveViewRatio</i>	11.886*** (0.517)	0.0003*** (0.000)	0.631*** (0.113)	2.459*** (0.422)
<i>Controls</i>	No	No	No	No
Observations	885 478	885 478	885 478	885 478
<i>Estimation</i>	OLS	OLS	Probit	Logit

Notes: *ActiveViewRatio* is measured during the initial 30-day exposure period following registration; dependent variables are measured during post-exposure month 1. The exposure and outcome windows do not overlap. Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E2: Long-Term Results Without Controls in Study 3

Panel A: Creator Productivity

Variable	(1) <i>NumUploadVideos</i>	(2) <i>ProducedVideoLength</i>	(3) <i>DaysOfProduction</i>
<i>ActiveViewRatio</i>	16.740*** (0.877)	868.094*** (52.122)	2.452*** (0.065)
<i>Controls</i>	No	No	No
Observations	885 478	885 478	885 478

Panel B: Creator Performance

Variable	(1) <i>NumCompleteViews</i>	(2) <i>NumLikes</i>	(3) <i>NumComments</i>	(4) <i>NumForwards</i>
<i>ActiveViewRatio</i>	76.724*** (4.018)	16.642*** (2.853)	6.428*** (1.135)	−0.431 (1.020)
<i>Controls</i>	No	No	No	No
Observations	885 478	885 478	885 478	885 478

Panel C: Creator Growth

Variable	(1) <i>NumNewFollowers</i>	(2) <i>MonetizeProbability</i>	(3) <i>MonetizeProbability</i>	(4) <i>MonetizeProbability</i>
<i>ActiveViewRatio</i>	1.202 (1.167)	0.001*** (0.000)	0.259*** (0.059)	0.872*** (0.208)
<i>Controls</i>	No	No	No	No
Observations	885 478	885 478	885 478	885 478
<i>Estimation</i>	OLS	OLS	Probit	Logit

Notes: *ActiveViewRatio* is measured during the initial 30-day exposure period following registration; dependent variables are aggregated over post-exposure months 1–6. The exposure and outcome windows do not overlap. Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E3: Short-Term Log-Transformation Results in Study 3

Panel A: Creator Productivity

Variable	(1) <i>NumUploadVideos</i> (log)	(2) <i>ProducedVideoLength</i> (log)	(3) <i>DaysOfProduction</i> (log)
<i>ActiveViewRatio</i>	0.191*** (0.004)	0.350*** (0.008)	0.161*** (0.003)
<i>Controls</i>	Yes	Yes	Yes
Observations	885 478	885 478	885 478

Panel B: Creator Performance

Variable	(1) <i>NumCompleteViews</i> (log)	(2) <i>NumLikes</i> (log)	(3) <i>NumComments</i> (log)	(4) <i>NumForwards</i> (log)
<i>ActiveViewRatio</i>	0.534*** (0.006)	0.419*** (0.006)	0.259*** (0.004)	0.100*** (0.003)
<i>Controls</i>	Yes	Yes	Yes	Yes
Observations	885 478	885 478	885 478	885 478

Panel C: Creator Growth

Variable	(1) <i>NumNewFollowers</i> (log)
<i>ActiveViewRatio</i>	0.497*** (0.010)
<i>Controls</i>	Yes
Observations	885 478

Notes: *ActiveViewRatio* is measured during the initial 30-day exposure period following registration; dependent variables are measured during post-exposure month 1. The exposure and outcome windows do not overlap. For each outcome Y , the log-transformed dependent variable is defined as $\ln(Y_i - \min_j\{Y_j\} + 1)$, where $\min_j\{Y_j\}$ is the sample minimum of that outcome in the corresponding outcome window. This shift retains observations for outcomes that can take negative values, including *NumNewFollowers*. Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *MonetizeProbability* is not included because it is a binary outcome.

Table E4: Long-Term Log-Transformation Results in Study 3

Panel A: Creator Productivity				
Variable	(1) <i>NumUploadVideos</i> (log)	(2) <i>ProducedVideoLength</i> (log)	(3) <i>DaysOfProduction</i> (log)	
<i>ActiveViewRatio</i>	0.166*** (0.006)	0.260*** (0.011)	0.150*** (0.005)	
<i>Controls</i>	Yes	Yes	Yes	
Observations	885 478	885 478	885 478	
Panel B: Creator Performance				
Variable	(1) <i>NumComplete</i> <i>Views</i> (log)	(2) <i>NumLikes</i> (log)	(3) <i>NumComments</i> (log)	(4) <i>NumForwards</i> (log)
<i>ActiveViewRatio</i>	0.526*** (0.009)	0.393*** (0.009)	0.267*** (0.007)	0.115*** (0.005)
<i>Controls</i>	Yes	Yes	Yes	Yes
Observations	885 478	885 478	885 478	885 478
Panel C: Creator Growth				
Variable	(1) <i>NumNewFollowers</i> (log)			
<i>ActiveViewRatio</i>	0.437*** (0.017)			
<i>Controls</i>	Yes			
Observations	885 478			

Notes: *ActiveViewRatio* is measured during the initial 30-day exposure period following registration; dependent variables are aggregated over post-exposure months 1–6. The exposure and outcome windows do not overlap. For each outcome Y , the log-transformed dependent variable is defined as $\ln(Y_i - \min_j \{Y_j\} + 1)$, where $\min_j \{Y_j\}$ is the sample minimum of that outcome in the corresponding outcome window. This shift retains observations for outcomes that can take negative values, including *NumNewFollowers*. Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *MonetizeProbability* is not included because it is a binary outcome.

Appendix F: Additional Dynamic Associations in Study 3

Figure F1 reports the remaining creator-productivity, performance, and growth outcomes. All plots use *ActiveViewRatio* measured during the initial 30-day exposure period and monthly dependent variables measured over post-exposure months 1–6, so the exposure and outcome windows do not overlap.

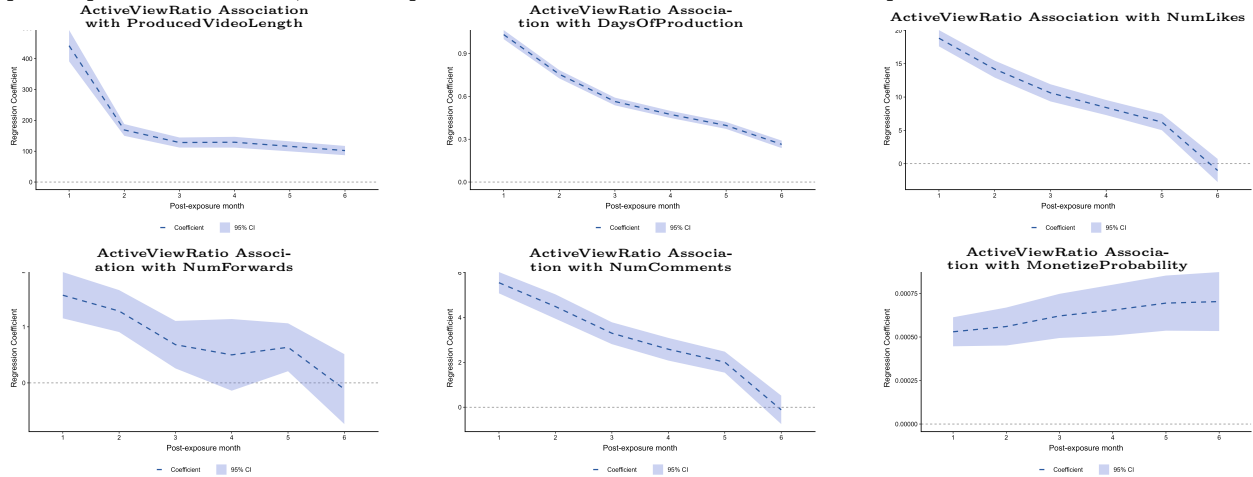


Figure F1: Additional Dynamic Associations with Creator Outcomes in Study 3

Notes. The dashed lines plot OLS coefficients from separate monthly regressions of the indicated outcome on *ActiveViewRatio* measured during the initial 30-day exposure period. The horizontal axis indexes post-exposure months 1–6, with post-exposure month 1 denoting the first month immediately following the initial exposure period. The shaded areas report 95% confidence intervals based on robust standard errors. All specifications control for gender, city level, region, and community type ($N = 885,478$ in each regression).

Appendix G: Suggestive Mediation Evidence

Having documented positive conditional associations between higher early active-viewer exposure share and long tail creators’ long-run outcomes, we examine whether the data are compatible with plausible engagement pathways. Drawing on prior literature (Oestreicher-Singer and Zalmanson 2013, Heinonen 2011), one possible explanation is that active viewers are more likely to engage with long tail creators’ content. Under this explanation, higher initial active-viewer exposure share would be associated with stronger engagement performance (likes, forwards, comments, and complete views), which would in turn be associated with long-term creator productivity, follower growth, and monetization potential.

We conduct a suggestive mediation decomposition using engagement performance metrics aggregated over post-exposure months 1–2 as intervening variables and creator outcomes in post-exposure months 3–5 as dependent variables. We estimate the decomposition with PROCESS Model 4 (Hayes 2022) on a random subsample of 100,000 creators (chosen for computational feasibility given the bootstrap intensity) with 5,000 bootstrap resamples. Results are presented in Table G1. Across productivity outcomes, the estimated total indirect associations through engagement metrics are positive, with bootstrap intervals excluding zero; for produced video length, the estimates through complete views and comments are positive. For growth outcomes, the estimated indirect association through forwards is positive for net new follower acquisition, while those through likes, comments, and complete views are positive for monetization probability. These decompositions are compatible with engagement-based pathways, but they do not establish causal mediation because neither active-viewer exposure share nor the intervening engagement measures are randomly assigned.

Table G1: Suggestive Mediation Decomposition of Engagement Performance and Creator Outcomes

Variable	Creator Productivity			Creator Growth	
	Videos Uploaded	Video Length	Prod. Days	New Followers	Monetization
Total Indirect Association	0.604	17.894	0.195	0.426	0.019
	[0.458, 0.784]	[11.456, 26.645]	[0.158, 0.237]	[0.051, 0.847]	[0.015, 0.024]
<i>Indirect associations via:</i>					
Likes	0.111	−2.941	0.070	0.158	0.008
	[0.005, 0.233]	[−9.177, 1.566]	[0.041, 0.103]	[−0.304, 0.699]	[0.005, 0.014]
Forwards	0.014	0.005	0.009	0.231	0.001
	[0.004, 0.033]	[−0.461, 0.552]	[0.003, 0.018]	[0.052, 0.471]	[−0.001, 0.001]
Comments	0.125	5.417	0.082	0.088	0.005
	[0.051, 0.212]	[0.064, 15.242]	[0.053, 0.122]	[−0.318, 0.551]	[0.002, 0.010]
Complete Views	0.353	15.413	0.033	−0.052	0.006
	[0.181, 0.588]	[7.834, 25.831]	[0.020, 0.049]	[−0.261, 0.141]	[0.003, 0.008]
Direct Association	4.292	286.039	0.765	−3.100	1.038
	[1.914, 6.670]	[69.601, 502.478]	[0.553, 0.976]	[−8.306, 2.106]	[−0.579, 2.654]
Total Association	4.896	303.933	0.961	−2.673	N/A
	[2.512, 7.281]	[87.456, 520.411]	[0.740, 1.181]	[−7.889, 2.542]	

Notes: Independent variable = *ActiveViewRatio*. Intervening variables are measured over calendar-relative Months 2–3 (post-exposure months 1–2); DVs are measured over calendar-relative Months 4–6 (post-exposure months 3–5). PROCESS Model 4 (Hayes, 2022), 5,000 bootstrap resamples on 100,000 creators. 95% percentile bootstrap CIs in brackets. *MonetizeProbability* uses logistic regression; total association not reported. Because neither exposure nor the intervening variables are randomized, the decomposition is descriptive and does not establish causal mediation.

Appendix H: Micro-Foundations, Proofs, and Derivations

This appendix micro-founds the primitives of the infinite-horizon model in Section 4 and collects all proofs and derivations. The proofs use Assumptions 1–3 only where explicitly invoked.

H.1. Domains, Feasibility, and Micro-Foundations

Let $\mathcal{X} = [0, \bar{x}]^2$ with $0 < \bar{x} < 1$, $a_j = \eta_j \Theta_j$, $\bar{Z} = (a_A + a_C)\bar{x}$, and $\bar{z} = \bar{Z}/N_L$. Since g is continuous on the compact set $[0, \bar{Z}] \times \mathcal{N}_L$, $G_{\max} = \max_{N_L \in \mathcal{N}_L} g(\bar{Z}, N_L)$ exists. Choosing $\mathcal{I} = [0, \bar{N}]$ with $\bar{N} \geq G_{\max}/\delta$ gives, for all feasible (N_K, x, N_L) ,

$$(1 - \delta)N_K + g(Z(x), N_L) \leq (1 - \delta)\bar{N} + G_{\max} \leq \bar{N}.$$

Thus the transition maps $\mathcal{I} \times \mathcal{X}$ into $\mathcal{I}$.

Engagement technology. A segment- j viewer derives per-slot utility

$$(1 - x)v(N_K) + x(v_L + b_j) - \frac{\bar{\gamma}}{2m(N_L)}x^2$$

from a feed with long tail share x , where the first term captures the utility from content by established creators, the second term captures that from content by long tail creators, and the last term captures the satiation from consuming long tail content. Collecting terms gives

$$v(N_K) - [v(N_K) - v_L - b_j]x - \frac{\bar{\gamma}}{2m(N_L)}x^2,$$

which is (5) with

$$c_j(N_K) = \Delta(N_K) - b_j, \quad \Delta(N_K) \equiv v(N_K) - v_L, \quad \gamma(N_L) = \bar{\gamma}/m(N_L). \quad (13)$$

The estimated sign pattern $c_A \leq 0 < c_C$ is equivalent to $b_C < \Delta(N_K) \leq b_A$: active viewers' diversity taste weakly exceeds the quality gap, while casual viewers' taste does not.

Development technology. Effective attention $Z = \eta_A \Theta_A x_A + \eta_C \Theta_C x_C$ is spread across the refreshed long tail breadth N_L , so the dose is $z = Z/N_L$. The perspective technology $g(Z, N_L) = N_L H(Z/N_L)$ satisfies

$$g_Z = H'(z) > 0, \quad g_{ZZ} = \frac{H''(z)}{N_L} < 0, \quad g_{N_L} = H(z) - zH'(z) > 0, \quad g_{ZN_L} = -\frac{zH''(z)}{N_L} > 0. \quad (14)$$

The sign of g_{N_L} follows from strict concavity and $H(0) = 0$: for $z > 0$, $H(z) > zH'(z)$; at $z = 0$, the derivative is zero. Because g is interpreted as effective creator capital rather than a literal fraction of creators promoted into a star category, no probability upper bound is required. If one uses a literal mass interpretation, we make the additional assumption $H([0, \bar{z}]) \subseteq [0, 1]$.

H.2. Bellman Existence, Monotonicity, and Concavity

Fix N_L and define

$$R(N_K, x; N_L) = \sum_{j \in \{A, C\}} \Theta_j \pi_j(x_j; N_K, N_L), \quad f(N_K, x; N_L) = (1 - \delta)N_K + g(Z(x), N_L).$$

The Bellman operator on $C(\mathcal{I})$ is

$$(\mathcal{T}U)(N_K) = \max_{x \in \mathcal{X}} \{R(N_K, x; N_L) + \beta U(f(N_K, x; N_L))\}.$$

Lemma. Under Assumption 1, $\mathcal{T}$ is a contraction with modulus β , has a unique continuous fixed point $V(\cdot; N_L)$, and this fixed point is strictly increasing and concave in N_K . It is locally Lipschitz on the interior of $\mathcal{I}$, so left and right derivatives exist at every interior point. No global differentiability is asserted.

Proof. Compactness and continuity imply that $\mathcal{T}$ maps $C(\mathcal{I})$ into itself by Berge's maximum theorem. For any continuous U_1, U_2 ,

$$\|\mathcal{T}U_1 - \mathcal{T}U_2\|_\infty \leq \beta \|U_1 - U_2\|_\infty,$$

so $\mathcal{T}$ is a contraction and has a unique continuous fixed point.

If U is nondecreasing, then $R(N_K, x; N_L)$ and $U(f(N_K, x; N_L))$ are nondecreasing in N_K , hence $\mathcal{T}U$ is nondecreasing. If U is nondecreasing and concave, $U \circ f$ is concave because f is increasing and concave in (N_K, x) . It remains to verify joint concavity of R . Writing

$$R(N_K, x; N_L) = \sum_j \Theta_j \left[(1 - x_j)v(N_K) + x_j(v_L + b_j) - \frac{\gamma(N_L)}{2} x_j^2 \right],$$

the Hessian in (N_K, x_A, x_C) has

$$R_{N_K N_K} = \left(\sum_j \Theta_j (1 - x_j) \right) v''(N_K), \quad R_{N_K x_j} = -\Theta_j v'(N_K), \quad R_{x_j x_j} = -\Theta_j \gamma(N_L),$$

and zero cross-partials across x_A and x_C . The x block is negative definite. Its Schur complement is nonpositive under the curvature condition in Assumption 1, so R is jointly concave. Maximizing a jointly concave function over the convex state-independent set $\mathcal{X}$ preserves concavity in N_K . Starting from $U_0 = 0$, the iterates $\mathcal{T}^n U_0$ are nondecreasing and concave, and uniform convergence preserves these properties.

Strict monotonicity follows because $\bar{x} < 1$ leaves positive established-content exposure for every feasible action:

$$R_{N_K}(N_K, x; N_L) = \left(\sum_j \Theta_j(1 - x_j) \right) v'(N_K) > 0.$$

For $N'_K > N_K$, evaluating at an optimizer for N_K gives a strictly larger current reward and a weakly larger continuation value. A finite concave function on an interval is locally Lipschitz in the interior, completing the proof. $\square$

H.3. Exact Dynamic Return

For fixed state N_K , define

$$D(Z; N_K, N_L) = \beta [V((1 - \delta)N_K + g(Z, N_L); N_L) - V((1 - \delta)N_K; N_L)], \quad Z \in [0, \bar{Z}].$$

Lemma. Under Assumption 1, $D(0; N_K, N_L) = 0$, and $D(\cdot; N_K, N_L)$ is continuous, strictly increasing, and strictly concave in Z .

Proof. The first statement follows from $g(0, N_L) = 0$. Continuity follows from continuity of V and g . Since $g_Z > 0$ and V is strictly increasing, $D(\cdot; N_K, N_L)$ is strictly increasing. For strict concavity, take $Z_1 \neq Z_2$ and $\lambda \in (0, 1)$. Let $Z_\lambda = \lambda Z_1 + (1 - \lambda)Z_2$ and $Y_i = (1 - \delta)N_K + g(Z_i, N_L)$. Strict concavity of g gives $Y_\lambda > \lambda Y_1 + (1 - \lambda)Y_2$. Since V is increasing and concave,

$$V(Y_\lambda; N_L) > \lambda V(Y_1; N_L) + (1 - \lambda)V(Y_2; N_L).$$

Multiplication by $\beta > 0$ proves strict concavity of $D(\cdot; N_K, N_L)$. $\square$

Because $D(\cdot; N_K, N_L)$ is concave, its one-sided derivatives exist on $(0, \bar{Z})$ and its superdifferential $\partial D(Z; N_K, N_L)$ is nonempty there. At boundaries we use the corresponding one-sided supergradients.

H.4. Target-and-Dispatch Reduction

Define the signed current opportunity cost

$$L(x; N_K, N_L) = \sum_j \Theta_j \left(c_j(N_K)x_j + \frac{\gamma(N_L)}{2}x_j^2 \right).$$

For $Z \in [0, \bar{Z}]$, the minimum dispatch cost is

$$C(Z; N_K, N_L) = \min_{\substack{x \in \mathcal{X} \\ a_A x_A + a_C x_C = Z}} L(x; N_K, N_L).$$

The feasible set is nonempty, compact, and convex; strict convexity of $L(\cdot; N_K, N_L)$ in x gives a unique minimizer $x^d(Z; N_K, N_L)$. Convexity of $C(\cdot; N_K, N_L)$ follows from the value function of a convex program with a linear equality constraint.

Since $L(\cdot; N_K, N_L)$ is a signed opportunity cost, its unconstrained minimum over target levels need not occur at zero. Let $Z^m(N_K, N_L) \in \arg \min_{Z \in [0, \bar{Z}]} C(Z; N_K, N_L)$ and define $\tilde{C}(Z; N_K, N_L) = C(Z; N_K, N_L) - C(Z^m; N_K, N_L)$. This nonnegative incremental cost is useful for interpretation; the policy and proof results below use the exact signed cost C .

The Bellman action problem is equivalent to

$$\max_{Z \in [0, \bar{Z}]} \{D(Z; N_K, N_L) - C(Z; N_K, N_L)\}.$$

Indeed, for any action x , the continuation payoff depends on x only through $Z = a_A x_A + a_C x_C$. Conditional on that Z , replacing x by the least-cost dispatch $x^d(Z)$ weakly improves current engagement and leaves the transition unchanged. Conversely, every feasible Z is generated by its dispatch. Since $D(\cdot; N_K, N_L)$ is strictly concave and $C(\cdot; N_K, N_L)$ is convex, the scalar objective is strictly concave and has a unique maximizer.

H.5. Proofs of Propositions 1 and 2

Proof. Bellman existence and shape follow from Appendix H.2. The scalar reduction in Appendix H.4 gives unique Z^* and $x^* = x^d(Z^*)$.

At an optimal target, concave optimality implies that there exists a shadow value $w^* \in \partial D(Z^*; N_K, N_L) \cap \partial C(Z^*; N_K, N_L)$, with one-sided interpretation at the boundary. The KKT conditions for the dispatch problem are, for each j ,

$$0 \in \Theta_j [c_j(N_K) + \gamma(N_L)x_j^*] - w^*a_j + N_{[0, \bar{x}]}(x_j^*),$$

where $N_{[0, \bar{x}]}(\cdot)$ is the normal cone of the interval. Dividing by Θ_j and using $a_j = \eta_j \Theta_j$ yields the projection formula in (11). A segment receives positive exposure only when $w^* \geq q_j(N_K)$, with strict inequality away from the lower-bound kink. Hence $q_A < q_C$ implies active traffic enters before casual traffic. When both shares are uncapped and interior, the normal-cone terms vanish and subtraction gives the displayed exposure difference.

For any fixed feasible target Z , the definition of C gives $L(x^d(Z); N_K, N_L) \leq L(x; N_K, N_L)$ for every feasible dispatch x that attains Z . Strict convexity of L makes $x^d(Z)$ unique, so equality holds only at $x = x^d(Z)$.

For Proposition 2, write

$$F(Z; N_K, N_L) = D(Z; N_K, N_L) - C(Z; N_K, N_L).$$

The function $F(\cdot; N_K, N_L)$ is strictly concave. Under Assumptions 2 and 3, the marginal dispatch cost is active-only until it reaches the casual effective-unit cost:

$$C'(Z; N_K, N_L) = q_A(N_K) + \frac{\gamma(N_L)Z}{\eta_A^2 \Theta_A}, \quad 0 < Z < Z^{\text{sw}}(N_K, N_L),$$

where $Z^{\text{sw}}(N_K, N_L) = \eta_A^2 \Theta_A [q_C(N_K) - q_A(N_K)] / \gamma(N_L)$. Hence $C'_+(0; N_K, N_L) = q_A(N_K)$ and $C'_+(Z^{\text{sw}}; N_K, N_L) = q_C(N_K)$.

Part (i) follows from the one-sided first-order condition at the lower boundary. The unique maximizer satisfies $Z^* = 0$ if and only if $F'_+(0; N_K, N_L) \leq 0$, which is equivalent to $d_0(N_K, N_L) \leq q_A(N_K)$. If $c_A(N_K) < 0$, then $q_A(N_K) < 0$ while $d_0(N_K, N_L) > 0$, so this inequality cannot hold and starvation is impossible.

Part (ii) covers the active-only region. If $d_0(N_K, N_L) > q_A(N_K)$, then $F'_+(0; N_K, N_L) > 0$, so the unique maximizer is positive. If, in addition, $D'_+(Z^{\text{sw}}; N_K, N_L) \leq q_C(N_K)$, then $F'_+(Z^{\text{sw}}; N_K, N_L) \leq 0$. Strict concavity therefore rules out an optimum strictly to the right of Z^{sw} , implying $0 < Z^* \leq Z^{\text{sw}}$. On this branch the dispatch constraint is $a_A x_A = Z^*$, so $x_A^* = Z^* / a_A > 0$ and $x_C^* = 0$. Conversely, if the optimum is in the incubate region, then $Z^* > 0$ implies $F'_+(0; N_K, N_L) > 0$, hence $d_0(N_K, N_L) > q_A(N_K)$. Because the optimum is not strictly to the right of Z^{sw} , concavity implies $F'_+(Z^{\text{sw}}; N_K, N_L) \leq 0$, or equivalently $D'_+(Z^{\text{sw}}; N_K, N_L) \leq q_C(N_K)$.

Part (iii) covers the boost region. If $d_0(N_K, N_L) > q_A(N_K)$ and $D'_+(Z^{\text{sw}}; N_K, N_L) > q_C(N_K)$, then $F'_+(Z^{\text{sw}}; N_K, N_L) > 0$, so strict concavity implies that the unique maximizer lies to the right of the switch: $Z^* > Z^{\text{sw}}$. Conversely, if $Z^* > Z^{\text{sw}}$, then $Z^* > 0$ implies $d_0(N_K, N_L) > q_A(N_K)$, and concavity implies $F'_+(Z^{\text{sw}}; N_K, N_L) > 0$, hence $D'_+(Z^{\text{sw}}; N_K, N_L) > q_C(N_K)$. For $Z^* > Z^{\text{sw}}$, the no-cap KKT system gives

$$a_A x_A^* = Z^{\text{sw}} + \frac{\eta_A^2 \Theta_A}{\eta_A^2 \Theta_A + \eta_C^2 \Theta_C} (Z^* - Z^{\text{sw}}), \quad a_C x_C^* = \frac{\eta_C^2 \Theta_C}{\eta_A^2 \Theta_A + \eta_C^2 \Theta_C} (Z^* - Z^{\text{sw}}),$$

so $x_A^* > 0$ and $x_C^* > 0$. □

H.6. Nonvacuity of the Assumptions

We demonstrate in this section that the assumption set is nonempty. For example, take $v(N) = a \log(1 + N)$, $m(N_L) = N_L / \bar{n}$, and any $H \in C^2([0, \bar{z}])$ with $H(0) = 0$, $H' > 0$, and $H'' < 0$. The curvature condition in Assumption 1 is satisfied uniformly if

$$\gamma(N_L) \geq \frac{\Theta a}{\sum_j \Theta_j (1 - \bar{x})} \quad \text{for all } N_L \in \mathcal{N}_L,$$

which follows from $v'(N)^2 / [-v''(N)] = a$ for the log specification. The routing order is satisfied by any parameters with $q_A(N_K) < q_C(N_K)$ on $\mathcal{I}$; in the equal-productivity baseline this is the cost ordering $c_A < c_C$.

H.7. Derivation of Remark 1

Proof. Suppose both segments start from the same baseline long tail share x^u , the experiment removes the same proportional share d of long tail exposure in each segment, and the outcome is measured in the engagement currency of π_j . Blocking moves the feed from x^u to $(1 - d)x^u$. Therefore

$$\Delta E_j = \pi_j((1 - d)x^u) - \pi_j(x^u) = d x^u \left[c_j + \frac{\gamma}{2}(2 - d)x^u \right],$$

which is (12). The attention-weighted average is positive under $\hat{c} > 0$ for the experimental allocation, and the segment contrast per exposed slot is $d(c_C - c_A)$. This is an interpretation of the Study 1 incidence pattern, not an identification theorem for the full dynamic model. □